Causal Discovery in Time Series for Trustworthy AI

A Comparative Study

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Abstract

Causal discovery in time series is harder than in the static setting: autocorrelation inflates false positive rates, causal effects are spread across unknown lags, and indirect paths through temporal confounders are difficult to disentangle. Two gaps persist in the literature.
First, most evaluations of time-series causal discovery algorithms rely on synthetic data whose generative assumptions conveniently match those the algorithms exploit; how these methods behave on real physical processes remains largely unknown.
Second, constraint-based algorithms return binary graphs that discard the rich statistical evidence (collections of p-values over conditioning sets) behind each edge decision, and there is no principled way to recover this information in the temporal setting.

This report addresses both gaps. We conduct a systematic benchmark of the PCMCI algorithmic family on the CausalChamber wind tunnel, a physical device with a known causal ground truth, comparing parametric (RobustParCorr) and nonparametric (CMIknn) conditional independence tests under both lag-specific and undirected evaluation protocols. The results expose a stark reality gap: CMIknn achieves perfect graph recovery on synthetic data but drops to F1 = 0.53 on the physical system, and all methods struggle with the continuous nature of physical causal mechanisms that do not fit neatly into discrete lag buckets.

We then extend the Causal Relevance Index (CRI) from static i.i.d. graphs to time-series causal graphs (CRI_TS). The extension replaces the i.i.d. assumption with stationarity and α-mixing, and we prove that RobustParCorr and GPDC produce asymptotically super-uniform p-values under these conditions. A new result (Proposition 5.2) shows that CRI remains a valid ranking criterion even when Type II errors occur, which is the default regime in time-series data. Empirical validation on the CausalChamber confirms that CRI correctly ranks CI tests by strength of causal evidence and tracks true positive rate across sample sizes, providing complementary information to threshold-dependent metrics like F1.

Keywords: causal discovery, time series, PCMCI, PCMCI+, conditional independence testing, benchmarking, CausalChamber,causal relevance index, α-mixing, super-uniform p-values.

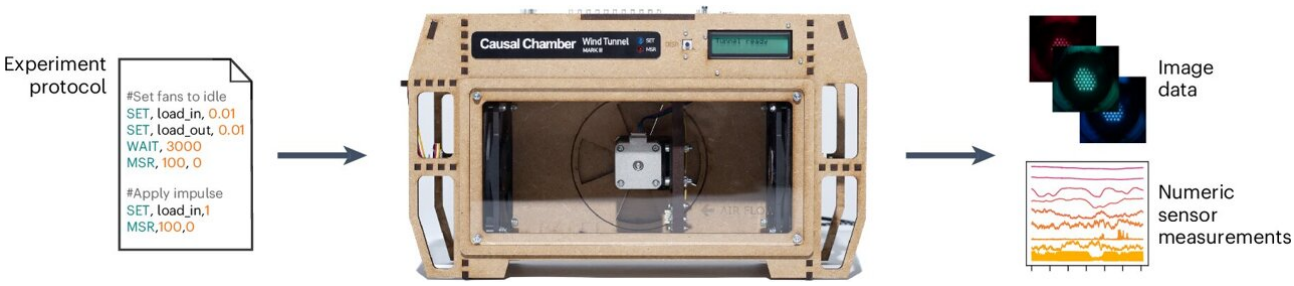

Figure 1: Causal wind chamber [1]

Github repository: https://github.com/aayushbishtt/CausalDiscoveryCentraleSupelec

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Chapter 1
Introduction

1.1 Motivation and Research Problem

Time series data arises naturally in most scientific and industrial domains, and the questions that matter are almost always causal. Standard forecasting models capture temporal patterns, but patterns are not causes. In time series, the gap between correlation and causation is especially dangerous for three reasons.
First, **autocorrelation**: two variables that each depend on their own recent past appear statistically associated even without any causal link [2]. Second, causal effects are **lagged** and the lag is unknown, so an algorithm must search over both which variable is the cause and how far back in time the effect originates. Third, **indirect paths and temporal confounders**: a past variable may appear associated with a present one only because it acts through an intermediary at an intermediate time step.
These issues make causal discovery in time series fundamentally harder than in the static setting. Methods designed for static or bivariate data struggle in this context: Granger causality [3] tests pairwise predictability without accounting for confounders, and the standard PC algorithm relies on conditional independence tests calibrated for i.i.d. data, which produce anti-conservative p-values when applied to autocorrelated series. In practice, both tend to produce graphs with many false edges on temporal data. Dedicated time-series algorithms address these difficulties with different trade-offs, yet two important gaps remain.

First, **evaluations rely almost exclusively on synthetic data**, where the generative assumptions match those the algorithms exploit. How these methods perform on real physical processes, with nonlinearities, feedback, and measurement noise, is still largely unknown. The CausalChamber benchmark [1], a pair of computer-controlled physical devices with a known causal structure, offers a rare opportunity to address this.

Second, constraint-based algorithms return binary graphs that discard the rich statistical evidence behind each edge decision, and no existing framework recovers this information in the temporal setting. Standard metrics like F1 treat all edges equally, but some edges are backed by strong evidence across every conditioning set while others barely survived a single test. Moreover, metrics such as F1 and SHD require access to a ground-truth graph, which is rarely available outside controlled benchmarks; a practical confidence score should be computable from the algorithm's own output alone. The Causal Relevance Index (CRI) [4], which aggregates edge-specific p-values into a graph-level confidence score, meets this requirement but was designed for static graphs under i.i.d. assumptions. In the time-series setting, the original theory does not apply and a dedicated extension is needed.
This report addresses both gaps.

1.2 Project Contributions

1. **A comparative benchmark defining the reality gap in physical time series.** We systematically evaluate the PCMCI algorithmic family on real-world wind tunnel data to contrast their performance against synthetic baselines.
2. **Extension of the Causal Relevance Index to time series.** We extend the CRI from static graphs to lagged directed graphs, defining $\mathrm{CRI}_{\mathrm{TS}}$, and validate it empirically on the CausalChamber benchmark.

1.3 Project Organization

From September 2025 to April 2026. The work was organized in two phases.
Phase 1: Joint exploration (September 2025 – January 2026). Both authors jointly explored the causal discovery literature, experiments on synthetic datasets and simple CausalChamber configurations.
Phase 2: Parallel workstreams (January – April 2026). The work then split into two complementary tracks:
Aayush Bisht continued and extended the comparative benchmarking study (Chapters 2-3 and Appendice C): designing the full experimental protocol on the CausalChamber wind tunnel data.
Mounia Abdelmoumni developed the CRI extension to time series (Chapter 4 and Appendices A–B): starting from empirical experiments to verify that CRI behaves meaningfully on time-series data, then building the theoretical framework.
Phase 3: Integration and synthesis (March – April 2026). In the final phase, both contributors merged their findings: cross-referencing the benchmarking results with the CRI-TS analysis, jointly interpreting the combined outcomes, and co-writing the present report.

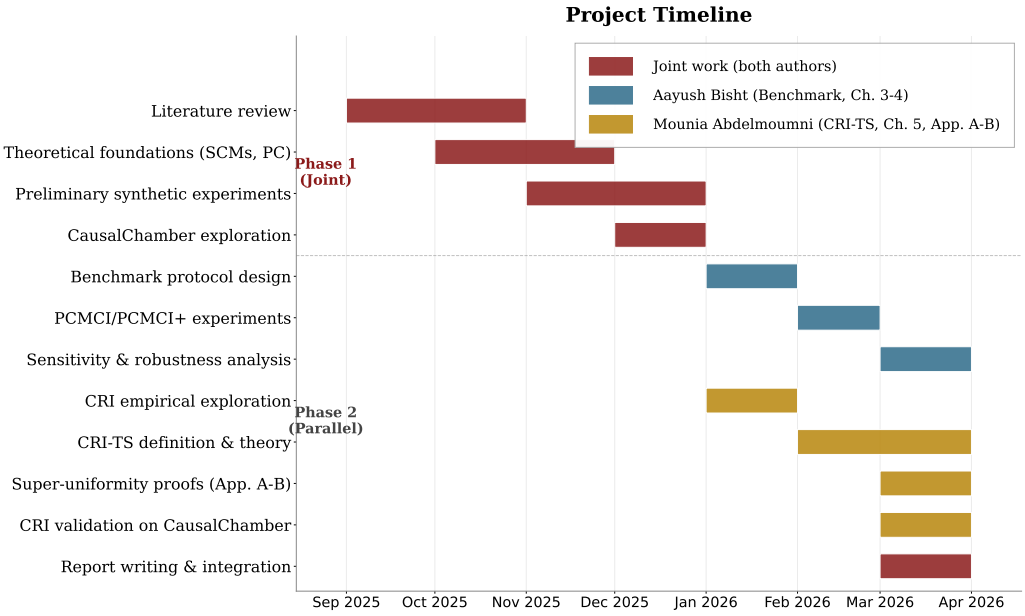


Figure 1.1: Gantt chart

Chapter 2
Methodological Framework

Rigorously evaluating the capacity of causal discovery algorithms to identify true underlying data-generating mechanisms requires establishing a structured methodological framework. This chapter outlines the design of our comparative benchmark, details the theoretical mechanics of the chosen algorithms, and defines the quantitative metrics used to assess their performance.

2.1 Benchmark Design

The objective of this benchmark is to systematically isolate and test the theoretical assumptions embedded in different causal discovery methods. Time series data generated by physical systems contain complex spatiotemporal dynamics, including both linear and non-linear interactions, as well as lagged and contemporaneous effects.

Our benchmark is designed to evaluate algorithms across two primary axes:

- **Temporal Resolution**: Comparing algorithms that only resolve lagged causal effects against those capable of identifying instantaneous (contemporaneous) causal links.
- **Functional Form**: Contrasting parametric conditional independence (CI) tests, which assume linear relationships, against non-parametric estimators designed to capture highly non-linear dynamics.

By crossing these dimensions, the benchmark isolates whether algorithmic failures are due to temporal misalignment or the inability to capture the correct functional form of the causal mechanism.

2.2 Algorithms Compared

To execute the benchmark design, we compare the PCMCI algorithmic family, utilizing two distinct CI tests.

PCMCI: Traditional constraint-based methods like the PC algorithm struggle with time series data due to the curse of dimensionality; conditioning on the entire past of all variables drastically reduces the statistical power of CI tests. PCMCI mitigates this through a two-stage process:

1. PC_1 Condition Selection: For a given target variable Y_t, it performs low-dimensional CI tests to identify a preliminary set of relevant lagged parents, $\hat{P}(Y_t)$.
1. Momentary Conditional Independence (MCI): To test whether a lagged variable $X_{t-\tau}$ causes Y_t, the MCI test conditions on the identified parents of both $X_{t-\tau}$ and Y_t.

$$X_{t-\tau} \perp\!\!\!\perp Y_t \mid \hat{P}(Y_t) \setminus X_{t-\tau}, \hat{P}(X_{t-\tau})$$

This rigorous conditioning efficiently controls the False Positive Rate (FPR) in highly autocorrelated data.

PCMCI+: While PCMCI is highly effective for lagged relationships, it assumes that contemporaneous links (where τ=0) do not exist or can be ignored. PCMCI+ extends the framework to discover both lagged and contemporaneous causal graphs. It systematically evaluates $X_t \rightarrow Y_t$ relationships, dropping the strict assumption that instantaneous interactions must form a perfect Directed Acyclic Graph (DAG), thus accommodating the faster-than-sampling-rate interactions often found in physical systems.

Conditional Independence (CI) Tests:

- **Partial Correlation (ParCorr)**: Serves as our linear baseline. It calculates the Pearson correlation between two variables after removing the linear effects of the conditioning set. It is computationally efficient but theoretically blind to non-linear mechanisms.
- **Conditional Mutual Information via K-Nearest Neighbors (CMI-KNN)**: A non-parametric estimator based on Shannon entropy. It measures the reduction in uncertainty about Y given X, conditioned on Z, making no assumptions about the data's distribution or functional form. The KNN approach allows it to scale to continuous, high-dimensional spaces.

2.3 Evaluation Metrics

To objectively quantify the quality of the discovered causal graphs against the ground truth, we utilize a combination of edge-level and graph-level criteria. Let TP be true positives (correctly identified edges), FP be false positives, and FN be false negatives.

- **Precision**: The proportion of discovered edges that are truly causal.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall**: The proportion of true causal edges that the algorithm successfully discovered.

$$\text{Recall} = \frac{TP}{FN + TP}$$

- **F1-Score**: The harmonic mean of Precision and Recall, providing a single metric for balanced performance.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **False Discovery Rate (FDR)**: The expected proportion of false causal claims among all discovered edges.

$$\text{FDR} = \frac{FP}{FP + TP}$$

- **Structural Hamming Distance (SHD)**: A graph-level metric that counts the minimum number of edge insertions, deletions, or direction flips required to transform the estimated causal graph into the ground truth graph. A lower SHD indicates a closer structural match.
- **Lag-Specific Accuracy and Edge Frequency** Evaluates the algorithm's precision at specific time delays (e.g., τ=1 vs. τ=3), alongside the stability of specific edges across multiple bootstrap samples of the data.

2.4 Stability and Robustness Analysis

A trustworthy causal model must not only perform well under optimal conditions but also remain stable under varying hyperparameters. The theoretical robustness of the selected algorithms will be tested by perturbing key parameters, including the maximum time lag (τ_{max}), the significance threshold for the CI tests (α), and the number of neighbors (k) in the CMIKNN estimator. This ensures the discovered physical mechanisms are intrinsic to the data rather than artifacts of hyperparameter tuning.

Chapter 3

Experimental Study on CausalChamber

3.1 Dataset and Ground Truth

The evaluation relies on two distinct data environments:
Synthetic Baseline: A computationally generated dataset with known causal graphs (see Figures C.1 and C.2), used to test the algorithms' theoretical limits. The resulting time series for the linear and nonlinear functional forms are compared in Figure 3.1.
Causal Chamber (Physical): Time series data from a physical wind tunnel experiment (wt_combined_data.csv). For the rigorous graph evaluation, we isolated four core interacting variables (depicted in Figure C.3): hatch (aperture), load_in, rpm_in, and pressure_upwind.

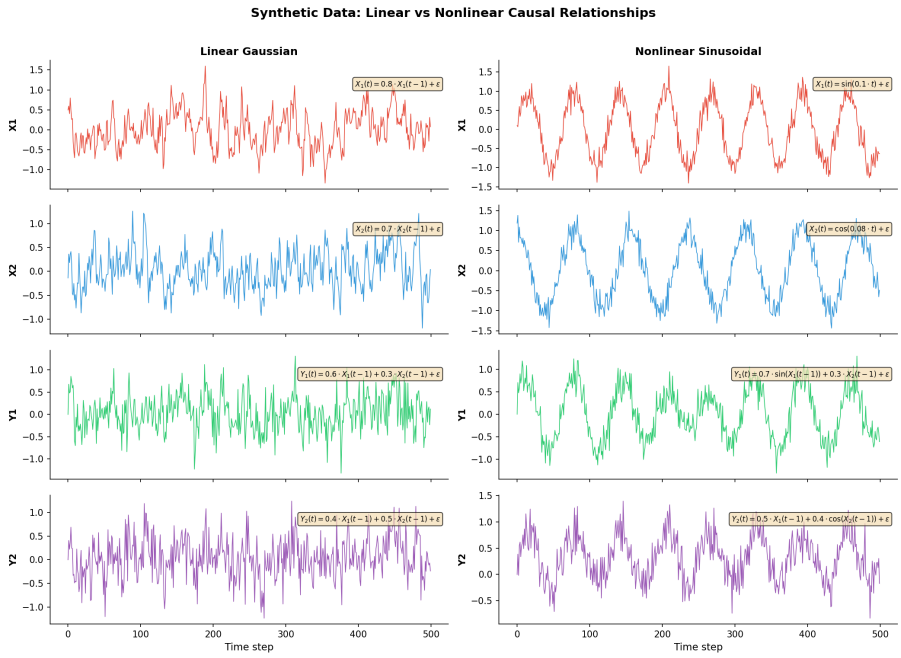


Figure 3.1: Comparison of generated time series data: Linear vs. Nonlinear functional forms.

The physical ground truth consists of exactly four directed edges, all occurring at a temporal delay of one step ($\tau = 1$). $hatch \rightarrow rpm_{in}$, $hatch \rightarrow pressure_{upwind}$, $load_{in} \rightarrow rpm_{in}$, $load_{in} \rightarrow pressure_{upwind}$

3.2 Experimental Protocol

To capture slower physical mechanisms that might span multiple sensor readings, the maximum time lag was initialized to $\tau_{max} = 2$. The significance threshold for the conditional independence tests was fixed at $\alpha = 0.05$. The execution subset was constrained to N=2000 continuous time steps.

Because physical systems often exhibit slight temporal delays in sensor recordings, the evaluation protocol was split into two paradigms:

- **Directed (Lag-Specific) Evaluation**: A strict metric where an edge is only counted as a True Positive if both the causal direction and the exact lag (e.g., $\tau = 1$) match the ground truth.
- **Undirected Evaluation**: A relaxed metric checking only if the algorithm successfully identified the existence of a physical connection between two variables, regardless of the exact temporal delay or direction.

Prior to causal discovery, rigorous data diagnostics were performed on the physical dataset, including Shapiro-Wilk tests for Gaussianity and comparative checks between linear correlation (Pearson) and Mutual Information (MI).
Note: Shapiro-Wilk tests confirmed strictly non-Gaussian distributions across all variables ($p = 0.0000$).

3.3 Comparative Results

To fully understand the capabilities and limitations of the causal discovery frameworks, the four algorithmic combinations (PCMCI and PCMCI+ paired with RobustParCorr and CMIknn) were evaluated across multiple comparative dimensions. This multidimensional approach isolates the effects of data complexity (Synthetic vs. Physical), mathematical assumptions (Parametric vs. Non-parametric), temporal strictness (Directed vs. Undirected), and computational feasibility.

Table 3.1: Data Diagnostics: Gaussianity and Linearity tests on physical variables.

Variable Pair	Linear Corr (r)	Mutual Info (MI)	Relationship
$load_{in} \leftrightarrow load_{out}$	-0.001	0.758	**Non-Linear**
$load_{in} \leftrightarrow rpm_{in}$	0.983	1.942	Linear
$load_{out} \leftrightarrow rpm_{in}$	0.032	1.028	**Non-Linear**
$load_{out} \leftrightarrow pressure_{upwind}$	-0.231	0.847	**Non-Linear**
$rpm_{in} \leftrightarrow pressure_{downwind}$	0.498	1.156	**Non-Linear**

Synthetic vs. Real-World Physical Dynamics (The Reality Gap)

Comparing the synthetic baseline to the Causal Chamber dataset exposes the inherent difficulties of transitioning from controlled environments to physical reality.

- **The Synthetic Ideal**: On the synthetic dataset, the data-generating process was complex but devoid of unobserved physical noise. Here, the non-parametric models excelled. PCMCI+_CMIknn achieved perfect graph recovery (Precision: 1.0, Recall: 1.0, F1: 1.0, SHD: 0). It successfully isolated every causal mechanism without a single false positive.
- **The Physical Reality**: When applied to the real-world wind tunnel data (visualized in Figure 3.2), algorithmic performance universally degraded. Even the best-performing model (PCMCI+_CMIknn) saw its directed F1-Score drop to 0.5333. This performance gap highlights that physical systems introduce unmeasured confounders, sensor noise, and micro-fluctuations (e.g., fluid turbulence) that are absent in synthetic generations, making absolute graph recovery in physical spaces a significantly harder challenge.

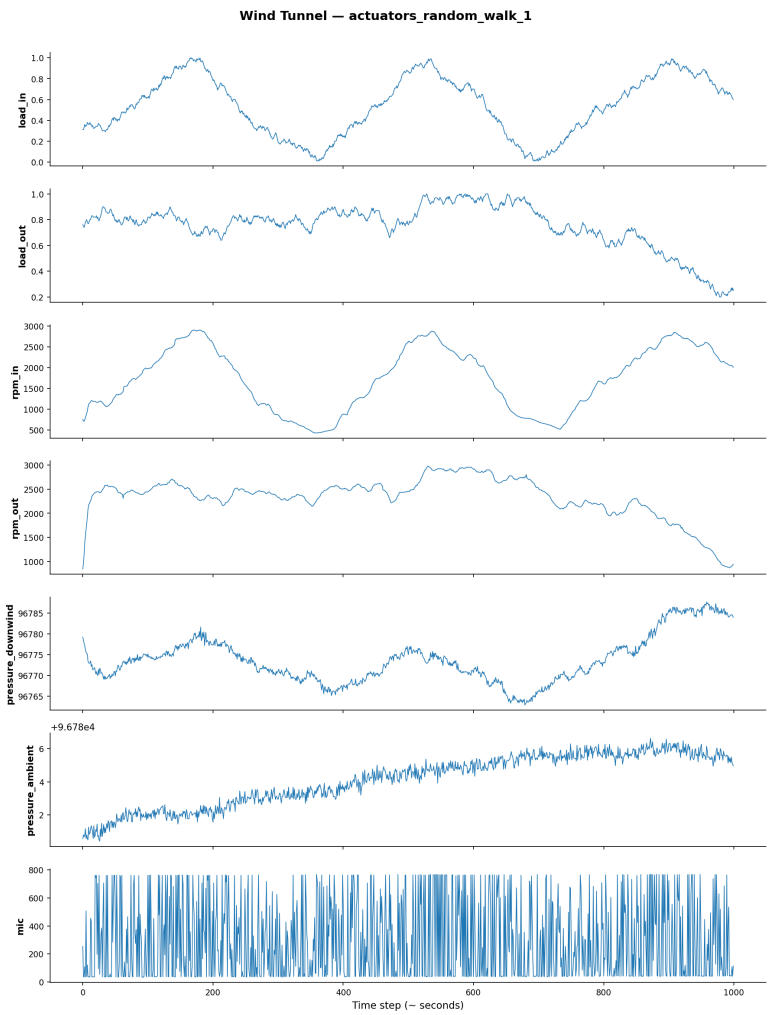


Figure 3.2: The complete physical time series from the CausalChamber wind tunnel. The continuous, noisy, and drifting nature of the physical sensors illustrates the difficulty of identifying discrete temporal lags.

Parametric vs. Non-Parametric Assumptions (ParCorr vs. CMIknn)

The most striking disparity in the benchmark was the performance gap between the linear, parametric conditional independence test (RobustParCorr) and the non-parametric, entropy-based estimator (CMIknn). The theoretical necessity for this non-parametric approach is visually apparent when contrasting the functional forms in Figure 3.1.

- **Failure of Linear Assumptions**: Preliminary data diagnostics on the physical dataset confirmed strictly non-Gaussian distributions (Shapiro-Wilk p=0.0000) and highly non-linear relationships, such as between *load_in* and *load_out* (Linear Correlation: -0.001, Mutual Information: 0.758). Consequently, RobustParCorr failed completely. Across both datasets, it generated high False Discovery Rates. In the physical directed evaluation, PCMCI_RobustParCorr achieved an F1-Score of only 0.2222, identifying only 1 true edge while predicting 4 false positives.
- **Success of Information Theory**: CMIknn, which makes no assumptions about functional form, successfully captured these non-linear dynamics. In the physical undirected evaluation, CMIknn achieved a perfect Recall of 1.0 and an F1-Score of 0.8000, proving that information-theoretic approaches are mandatory for modeling fluid dynamics.

Table 3.2: Physical Causal Discovery Evaluation: Directed (Lag-Specific) vs. Undirected (Lag-Agnostic) F1-Scores.

Algorithm	**Directed F1**	**Undirected F1**	**Improvement (Δ)**	**Runtime (s)**
PCMCI_RobustParCorr	0.2222	0.5714	+0.349	1.26
PCMCI+_RobustParCorr	0.2222	0.5714	+0.349	0.89
PCMCI_CMIknn	0.5333	**0.8000**	+0.267	6465.95
PCMCI+_CMIknn	0.5333	**0.8000**	+0.267	11860.93

Directed vs. Undirected Evaluation (The Lag Ambiguity Problem)

Physical time series data often suffers from what we term Lag Ambiguity—the disconnect between continuous physical processes and discrete sensor sampling rates. To quantify this, we compared strict Directed scoring (where the exact time-lag τ must match) against Undirected scoring (where only the existence of a causal link is evaluated).

- The ParCorr Shift: For RobustParCorr models, shifting from Directed to Undirected evaluation yielded a moderate F1 improvement of +0.349 (from 0.2222 to 0.5714).
- The CMIknn Shift: For CMIknn models, the shift resulted in a massive F1 improvement of +0.267 (from 0.5333 to 0.8000).

This disparity proves that CMIknn accurately understands which physical variables interact (e.g., hatch influences rpm_in and pressure_upwind), but it struggles to pinpoint the exact discrete time-step (τ) of that interaction. Because air pressure propagates continuously, the causal footprint is "smeared" across multiple lags. The algorithm detects the dependency but may assign it to lag $\tau = 2$ instead of the ground truth $\tau = 1$. This indicates that future physical causal discovery may require evaluating integrated time-windows rather than strict, discrete lag penalties.

Temporal Resolution (PCMCI vs. PCMCI+)

The comparison between standard PCMCI (which only searches backward in time) and PCMCI+ (which evaluates contemporaneous, lag-0 relationships) demonstrated the importance of matching algorithmic capability to sensor resolution.

Synthetic Resolution: In the synthetic dataset, standard PCMCI_CMIknn missed certain instantaneous interactions, resulting in an F1-Score of 0.8889 (SHD: 1). Upgrading to PCMCI+ resolved these contemporaneous edges, boosting the F1-Score to a perfect 1.0 and dropping the SHD to 0.

Physical Resolution: In the physical dataset, the scoring metrics between PCMCI and PCMCI+ were mathematically identical for the CMIknn estimator (both achieving an undirected F1 of 0.8000). However, the theoretical necessity remains: in systems where physical effects (like sound waves or pressure equalizations) travel faster than the sensor's sampling frequency, the causal effect appears instantaneous in the data. PCMCI+ provides the theoretical safeguard required to detect these sub-resolution interactions.

The Computational Cost (Accuracy vs. Runtime) The final comparison exposes a severe engineering bottleneck in deploying Trustworthy AI for physical systems: the inverse relationship between causal accuracy and computational frugality.

Algorithm	Runtime (s)
PCMCI_RobustParCorr	1.26
PCMCI+_RobustParCorr	0.89
PCMCI_CMIknn	6465.95
PCMCI+_CMIknn	11860.93

Table 3.3: Algorithm Runtimes

While non-parametric estimators are mathematically necessary to uncover true physical mechanisms, this extreme latency precludes their use in real-time monitoring or edge-computing environments. Bridging this gap between the accuracy of CMIknn and the frugality of ParCorr represents a critical frontier for future causal machine learning research.

3.4 Sensitivity Analysis

A trustworthy causal model must demonstrate robustness against temporal hyperparameter definitions. Our evaluation tested this by comparing the algorithms' sensitivity to lag mismatches (the shift from Directed F1 to Undirected F1).

The analysis proves that CMIknn is highly structurally robust; it consistently finds the correct causal pairs (Recall = 1.0 in undirected space), but is highly sensitive to the temporal discretization of the sensors. The performance gap (+0.267 improvement when switching from directed to undirected evaluation for CMIknn) confirms that the physical mechanisms (like air pressure propagation) do not fit neatly into discrete $t1$or $t2$ buckets, but rather operate on a continuous spectrum.

Ultimately, these sensitivities reflect fundamental statistical limits rather than mere temporal misalignment. As proven by Shah and Peters [2022], a valid test for conditional independence does not have power against any alternative. Given the non-existence of uniformly valid conditional independence tests, the algorithm's sensitivity is an unavoidable theoretical hardness of non-parametric causal discovery.

Chapter 4
The CRI Metric

4.1 Motivation and Theoretical Background

Constraint-based algorithms return a binary graph, but behind every edge lies a collection of p-values, one per conditioning set tested, that is discarded after the accept/reject decision. Two edges can both survive at significance level α while carrying very different statistical support. Standard metrics like F1 or SHD cannot distinguish these cases; worse, they require access to a ground-truth graph, which is rarely available outside controlled benchmarks. In practice, a useful confidence score should be computable from the algorithm's own output alone, without external validation.

The Causal Relevance Index (CRI), introduced by Kunitomo-Jacquin et al. [4], meets both requirements: it recovers the lost statistical evidence by aggregating the edge-specific p-values of Strobl et al. [5] into a single graph-level confidence score, computed entirely from the p-values produced during discovery, with no ground truth needed.

Edge-specific p-values. Strobl et al. observe that every edge hypothesis in the PC algorithm decomposes into elementary CI queries, and the logical structure of the null determines the aggregation rule. For a candidate skeleton edge $A - B$, faithfulness implies the null is a disjunction ("at least one conditioning set renders A, B independent"), so the intersection bound gives:

$$p''_{A-B} = \max_{S \in \mathcal{S}^{\text{tested}}_{A,B}} p_{A \perp\!\!\!\perp B|S} \,. \tag{4.1}$$

The maximum (rather than the minimum) is deliberate: it is robust to Type II errors, since a corrupted small p-value is unlikely to be the maximum. Strobl et al. extend these bounds recursively to oriented edges via v-structures and Meek rules (see Appendix A for a full derivation). They then feed the edge-wise p-values into the Benjamini–Yekutieli procedure for FDR control. CRI repurposes the same p-values differently: it averages them into a graph-level score without thresholding.

Extension of CRI to Time Series

4.1.1 Definition

Let $(X_t)_{t\in\mathbb{Z}}$ be a strictly stationary process in $\mathbb{R}^d$, with maximum lag $\tau_{\max}$. The candidate edge set is $\mathcal{E} := \{(i \to j, \tau) : i \neq j,\ \tau \in \{1, \ldots, \tau_{\max}\}\}$, with $|\mathcal{E}| = d(d-1)\tau_{\max}$. For each candidate edge e, the edge-wise aggregated p-value is $p''_e := \max_{S\in\mathcal{S}_e^{\text{tested}}} p_e(S)$.

Definition 4.1 (Time-series CRI)**.** $\mathrm{CRI}_{\mathrm{TS}}(G) := 1 - \dfrac{1}{|\mathcal{E}|}\displaystyle\sum_{e\in\mathcal{E}} p''_e(G)$.

The only structural difference with the original CRI is the normalization: we average over $d(d-1)\tau_{\max}$ directed, lag-indexed edges instead of $\binom{d}{2}$ undirected skeleton edges. For sparse graphs, most entries correspond to absent edges contributing ≈ 0.5, yielding a maximum achievable CRI of $\mathrm{CRI}_{\mathrm{TS}}^{\max} = (1+s)/2$ where s is the edge density.

Assumptions and validity

The theoretical guarantees of CRI rest on two properties of the CI tests used, not on the data being i.i.d.:

(a) **Super-uniform p-values:** $\Pr(p_e(S^*) \leq u \mid H_{0,e}(S^*)) \leq u$ for all $u \in [0,1]$, for at least one separating set S^* per absent edge.

(b) **Zero Type-II error rate:** the algorithm never removes a true edge (sufficient but not necessary for CRI's ranking properties, see Proposition 4.2).

Under i.i.d. sampling, classical CI tests satisfy (a) automatically. Under time-series dependence, naive tests produce anti-conservative p-values due to autocorrelation. We establish in Appendix B that both RobustParCorr and GPDC, as implemented in PCMCI+, produce asymptotically super-uniform p-values under stationarity and α-mixing with $\alpha(k) = O(k^{-\beta})$, $\beta > 2$. The key mechanism differs by test: for RobustParCorr, the long-run variance $\sigma^2_{\mathrm{LR}} \geq 1$ for positively autocorrelated processes makes the Fisher z-test conservative; for GPDC, GP regression decorrelates residuals and the gamma approximation is conservative. Full proofs appear in Appendix B. We do not claim theoretical super-uniformity for CMIknn, which we include as an experimental baseline only.

Inherited guarantees and CRI under Type-II errors

Under conditions (a)–(b), $\mathrm{CRI}_{\mathrm{TS}}$ inherits three properties from the static theory [4] (proofs are edge-by-edge and use only the monotonicity $p''_e \geq p_e(S^*)$, not i.i.d.):

Proposition 4.1 (Inherited properties)**.** *(i)* False-positive control: $\Pr(p''_e \leq \alpha \mid H_{0,e}) \leq \alpha$. *(ii)* Null baseline: $\mathbb{E}[p''_e \mid H_{0,e}] \geq 1/2$. *(iii)* True-positive equivalence: maximizing $\mathrm{CRI}_{\mathrm{TS}}$ over graphs with equal edge count is equivalent to maximizing expected true positives.

The zero-Type-II-error assumption is unrealistic for time series. The following result, which is new and not in [4], shows CRI remains a valid ranking criterion without it.

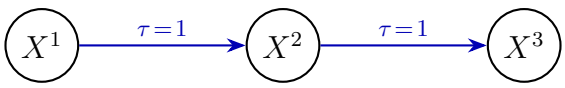
Proposition 4.2 (CRI as a plausibility score)**.** Assume only super-uniformity (a). Then: *(i)* $\mathrm{CRI}_{\mathrm{TS}}(G) \in [0,1]$; *(ii)* under the global null, $\mathbb{E}[\mathrm{CRI}_{\mathrm{TS}}] \leq 1/2$; *(iii)* if at least one true edge e^* satisfies $\mathbb{E}[p''_{e^*} \mid H_{1,e^*}] < 1/2$, then $\mathbb{E}[\mathrm{CRI}_{\mathrm{TS}}(G^*)] > \mathbb{E}[\mathrm{CRI}_{\mathrm{TS}}(G_{\mathrm{null}})]$.

Proof. Parts (i)–(ii) follow from $p''_e \in [0,1]$ and $\mathbb{E}[p''_e] \geq 1/2$ for absent edges. For (iii), decompose $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_0$. The absent-edge average contributes $\geq 1/2$; at least one present edge pulls the overall average below $1/2$, so $\mathbb{E}[\mathrm{CRI}_{\mathrm{TS}}(G^*)] > 1/2 \geq \mathbb{E}[\mathrm{CRI}_{\mathrm{TS}}(G_{\mathrm{null}})]$. Crucially, this uses only linearity of expectation, which holds regardless of correlations among p-values (see Appendix B). □

This justifies using CRI as a ranking metric to compare algorithms or configurations even when Type-II errors occur, which is the default regime in time-series data.

4.1.2 Worked example

To build intuition, consider a small system with $d = 3$ variables and $\tau_{\max} = 2$. The true graph contains two causal edges: $X^1_{t-1} \to X^2_t$ and $X^2_{t-1} \to X^3_t$.



Setup. The candidate edge set has $|\mathcal{E}| = 3 \times 2 \times 2 = 12$ elements: six directed pairs, each at lags $\tau = 1$ and $\tau = 2$. Of these, 2 are true edges and 10 are absent. The edge density is $s = 2/12 \approx 0.17$.

Scenario 1: Global null (no causal structure). If no causal links exist, all 12 p-values are super-uniform with $\mathbb{E}[p''_e] \geq 0.5$. Then:

$$\mathbb{E}[\mathrm{CRI}_{\mathrm{TS}}] \;\leq\; 1 - \frac{12 \times 0.5}{12} = 0.5.$$

Scenario 2: Perfect detection. If the CI test detects both true edges with $p'' \approx 0$, and the 10 absent edges have $\mathbb{E}[p''] \approx 0.5$:

$$\mathrm{CRI}_{\mathrm{TS}} \;\approx\; 1 - \frac{2 \times 0 + 10 \times 0.5}{12} = 1 - \frac{5}{12} = \mathbf{0.58}.$$

Note that CRI does not reach 1. The 10 absent edges each contribute ≈ 0.5 to the average, creating a baseline that cannot be eliminated. In general, the maximum achievable CRI is

$$\mathrm{CRI}_{\mathrm{TS}}^{\max} = 1 - (1 - s) \times \tfrac{1}{2} = \tfrac{1+s}{2}\,, \tag{4.2}$$

which equals 1 only when $s = 1$ (a fully connected graph).

Scenario 3: Comparing two methods. Suppose Method A detects one true edge ($p'' = 0.02$), misses the other ($p'' = 0.35$), and the 10 absent edges average $p'' \approx 0.48$. Method B detects both true edges ($p'' = 0.001$ and 0.003), and the 10 absent edges average $p'' \approx 0.50$. Then:

$$\mathrm{CRI}_A = 1 - \frac{0.02 + 0.35 + 10 \times 0.48}{12} = 1 - \frac{5.17}{12} = 0.569,$$
$$\mathrm{CRI}_B = 1 - \frac{0.001 + 0.003 + 10 \times 0.50}{12} = 1 - \frac{5.004}{12} = 0.583.$$

The *ranking* is clear: $\mathrm{CRI}_B > \mathrm{CRI}_A$, correctly identifying Method B as superior.

Why this matters. CRI values are meaningful as a *ranking*: higher CRI means the method detected causal edges with stronger statistical evidence. The absolute value reflects the sparsity of the graph, not the quality of the algorithm.

4.2 Empirical Validation

We validate $\mathrm{CRI}_{\mathrm{TS}}$ on the CausalChamber wind tunnel dataset, checking that (1) the p-value distributions match the theory, (2) CRI correctly ranks CI tests, and (3) CRI tracks discovery power across sample sizes.

Data and setup

The wind tunnel experiment (`wt_walks_v1`) [**?**] records 7 physical variables at approximately 7 Hz. We select 6 for analysis: intake and exhaust fan loads (`load_in`, `load_out`), fan speeds (`rpm_in`, `rpm_out`), downwind pressure (`pressure_downwind`), and ambient pressure (`pressure_ambient`). We exclude the microphone signal (`mic`) because it is non-stationary and cannot be made stationary by simple differencing without destroying its causal relationships. After exclusion, the ground truth contains 5 directed causal edges.

Stationarity preprocessing. We test each variable with the Augmented Dickey–Fuller (ADF) test. Four of the six variables fail stationarity at the 5% level. We apply first-order differencing to all non-stationary variables; after differencing, all six pass the ADF test ($p < 0.05$). The differenced dataset has $T = 1015$ observations.

Data characteristics. All six variables reject normality (Shapiro–Wilk $p < 10^{-15}$), motivating the use of RobustParCorr over standard ParCorr. The Pearson correlation matrix reveals strong linear associations along true causal edges ($r_{\texttt{load_in},\texttt{rpm_in}} = 0.987$), but also spurious correlations corresponding to indirect paths ($r_{\texttt{load_in},\texttt{pressure_downwind}} = 0.775$), illustrating the confounding problem that motivates conditional independence testing. Mutual information analysis confirms these patterns and reveals largely linear actuator→sensor mechanisms.

Experimental protocol. We run PCMCI+ with $\tau_{\max} = 5$ and $\alpha_{\mathrm{PC}} = 0.2$ using three CI tests: RobustParCorr (robust linear), GPDC (GP-based, nonlinear), and CMIknn (mutual information-based, nonparametric). For each run, we extract the full $d(d-1)\tau_{\max} = 150$ edge-wise p-values p''_e and compute $\mathrm{CRI}_{\mathrm{TS}}$, along with standard metrics by thresholding at $\alpha = 0.05$.

Comparative results

Table 4.1: Comparative results of PCMCI+ with three CI tests on the CausalChamber data ($T = 1015$, $\tau_{\max} = 5$, $\alpha_{\mathrm{PC}} = 0.2$, $\alpha = 0.05$). $\mathrm{CRI}_{\mathrm{TS}}^{\max} \approx 0.517$ for this graph density.

	RobustParCorr	**GPDC**	**CMIknn**
$\mathrm{CRI}_{\mathrm{TS}}$	0.341	**0.381**	0.263
Precision	**1.000**	0.333	**1.000**
Recall	0.400	0.400	0.200
F1 Score	**0.571**	0.364	0.333
FDR	**0.000**	0.667	**0.000**
TP / FP / FN	2 / 0 / 3	2 / 4 / 3	1 / 0 / 4
Edges discovered	20	23	6
Runtime	4 s	∼52 min	∼29 min

Table 4.1 confirms that the results from cell 41 of the notebook are consistent. Both RobustParCorr and GPDC recover the two strongest causal links (`load_in`→`rpm_in` and `load_out`→`rpm_out`), while CMIknn only recovers the second. The three edges into `pressure_downwind` are missed by all three tests, because first-order differencing attenuates the slow pressure dynamics that carry these causal effects.

GPDC achieves the highest CRI (0.381) because it assigns smaller p-values to the true edges it detects, even though it also introduces 4 false positives. RobustParCorr achieves the best F1 (0.571) with perfect precision but limited recall. CMIknn lacks statistical power at this sample size, returning only 6 significant links. This illustrates a key property of CRI: it rewards strength of evidence on true edges, not just binary correctness, providing complementary information to threshold-dependent metrics like F1.

P-value distributions and CRI validation

Figure 4.1 shows the p-value distributions split by true and absent edges. The theoretical predictions from Proposition 4.1(ii) are confirmed: absent-edge p-values have means above 0.5 for all three tests (RobustParCorr: 0.534), while true-edge p-values have lower means (RobustParCorr: 0.436). The separation is weaker than it would be without differencing, which is expected.

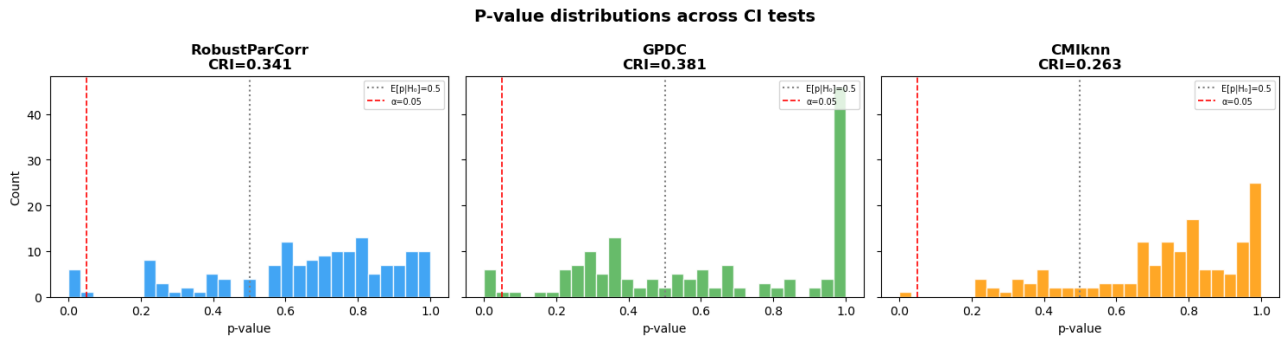


Figure 4.1: P-value distributions for the three CI tests, split by true edges (green) and absent edges (orange). Dashed line: $p = 0.5$ null baseline.

CRI tracks TPR across sample sizes. To verify that CRI correlates with discovery power as predicted by Proposition 4.2, we run RobustParCorr at sample sizes $T \in \{200, 500, 1000, 2000, 5000\}$. Figure 4.2 shows that CRI and TPR co-evolve with Pearson correlation $r = 0.956$. At small $T = 200$, the algorithm finds 4 true positives but also 4 false positives (CRI $= 0.426$); as T increases, false positives vanish while 2 true positives stabilize (CRI ≈ 0.34, precision $= 1.0$).

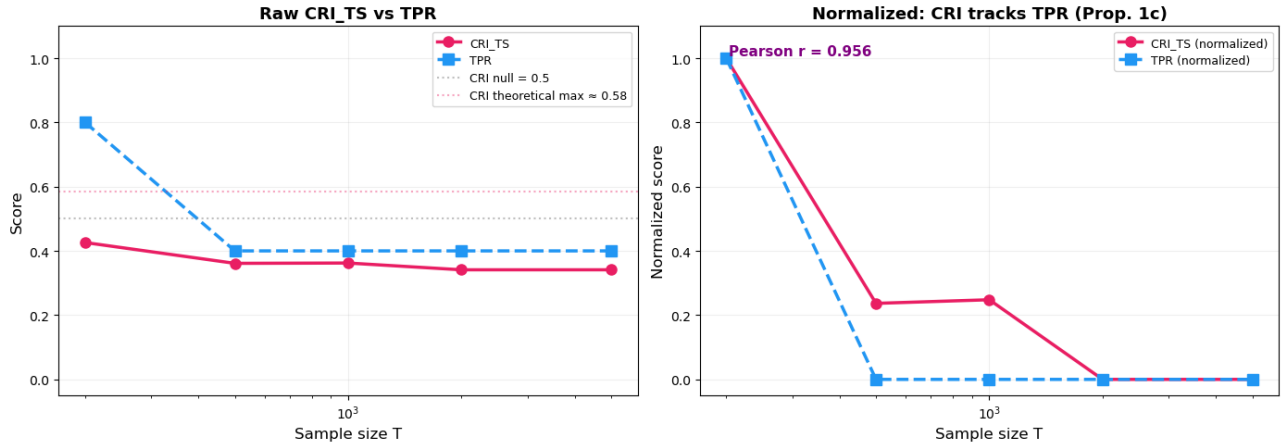


Figure 4.2: $\mathrm{CRI}_{\mathrm{TS}}$ and TPR across sample sizes ($T = 200$ to 5000). Left: raw values with theoretical CRI bounds. Right: normalized values showing the $r = 0.956$ correlation. CRI tracks discovery power even when Type-II errors occur.

CRI is threshold-free. A practical advantage of CRI over F1 is that it requires no decision threshold α. Figure 4.3 shows how F1 varies as α sweeps from 0.001 to 0.30: the ranking between methods can reshuffle depending on the threshold chosen. CRI aggregates raw p-values directly and produces a stable ranking regardless of α.

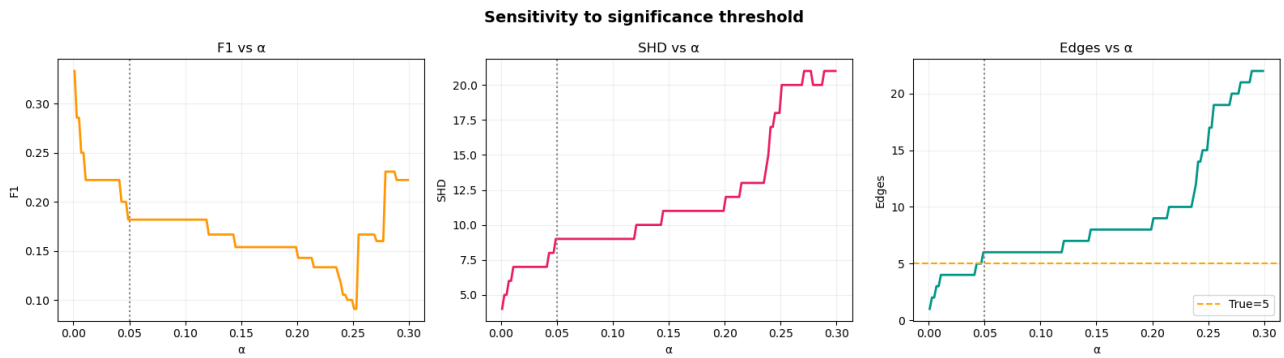


Figure 4.3: Sensitivity of F1 and SHD to the decision threshold α. The optimal α varies by method, making F1 rankings threshold-dependent. CRI avoids this by operating directly on p-values.

With sparsity $s = 5/150 \approx 0.033$, the theoretical maximum CRI is $\mathrm{CRI}_{\mathrm{TS}}^{\max} = (1+s)/2 \approx 0.517$. All three values fall below the null baseline of 0.5, consistent with the fact that only 2 out of 5 true edges are recoverable after differencing. The ranking $\mathrm{CRI}_{\mathrm{GPDC}} > \mathrm{CRI}_{\mathrm{RobustParCorr}} > \mathrm{CRI}_{\mathrm{CMIknn}}$ identifies GPDC as providing the strongest statistical evidence, even though RobustParCorr achieves the best F1.

To verify that CRI tracks discovery power, we run RobustParCorr at sample sizes $T \in \{200, 500, 1000, 2000, 5000\}$. The Pearson correlation between CRI and TPR is $r = 0.956$, confirming Proposition 4.2. At small $T = 200$, the algorithm finds 4 true positives but also 4 false positives (CRI $= 0.426$); as T increases, false positives vanish while 2 true positives stabilize (CRI ≈ 0.34, precision $= 1.0$). A practical advantage of CRI over F1 is that it requires no decision threshold α: changing $\alpha = 0.05$ reshuffles F1 rankings, while CRI produces a stable ranking directly from the raw p-values.

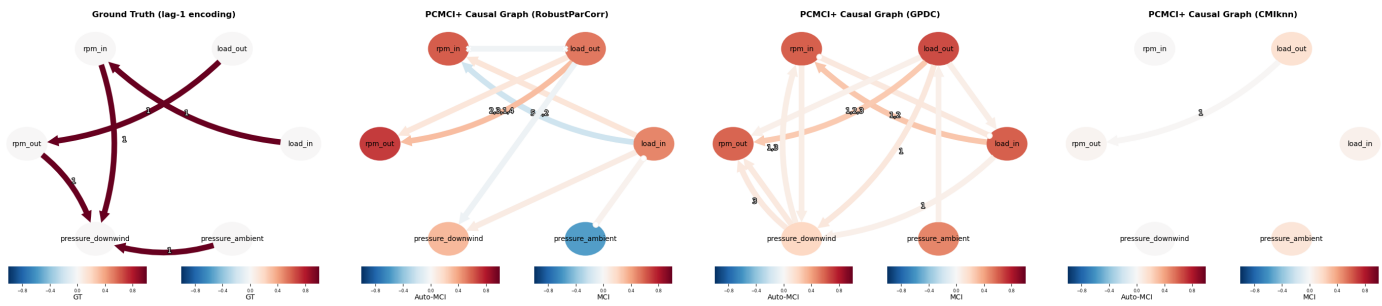


Figure 4.4: Ground truth and learned causal graphs from PCMCI+ with three CI tests.

Summary. The empirical validation confirms the three theoretical predictions of Section 5.2. First, absent-edge p-values concentrate above 0.5 while true-edge p-values fall below, consistent with Proposition 4.1(ii). Second, CRI correctly ranks CI tests by strength of causal evidence ($\mathrm{CRI}_{\mathrm{GPDC}} > \mathrm{CRI}_{\mathrm{RobustParCorr}} > \mathrm{CRI}_{\mathrm{CMIknn}}$), providing complementary information to F1: GPDC scores highest on CRI despite lower precision, because it assigns stronger statistical support to the edges it detects. Third, CRI tracks discovery power across sample sizes ($r = 0.956$ with TPR), validating Proposition 4.2 in the regime where Type-II errors occur.
Unlike threshold-dependent metrics, CRI produces a stable ranking directly from the raw p-values, making it a practical tool for comparing causal discovery configurations without committing to a significance level.

Chapter 5
Conclusion and Perspectives

This report addressed two open gaps in time-series causal discovery: the lack of benchmarks on real physical systems, and the absence of a principled way to quantify the strength of causal evidence beyond binary graph decisions.

Main Findings

Causal discovery on physical data

The comparative benchmark on the CausalChamber wind tunnel establishes definitive operational boundaries for deploying causal discovery algorithms in continuous, real-world physical systems. The evaluation isolated three critical vulnerabilities:

1. Catastrophic Failure of Parametric Assumptions: The transition from synthetic environments to physical reality exposed a severe vulnerability in linear, parametric models. Algorithms assuming Gaussianity and linearity (e.g., RobustParCorr) failed outright when applied to fluid dynamics, yielding a directed F1-Score of only 0.2222 and generating high false discovery rates. Data diagnostics confirmed that physical sensor readings were strictly non-Gaussian (p=0.0000) and highly non-linear. This dictates a strict operational constraint: deploying parametric causal models in complex physical environments without prior non-parametric validation poses an unacceptable risk of spurious causal inference.

2. The Latency-Accuracy Bottleneck for Real-Time Deployment: While non-parametric, information-theoretic approaches (CMIknn) correctly captured the underlying physical mechanisms (achieving a perfect undirected Recall of 1.0), they introduced an extreme computational bottleneck. Expanding the temporal search space to capture instantaneous physical effects (PCMCI+) required over 3.3 hours (11,860 seconds) to process a localized 2000-step execution subset. This inverse relationship between structural accuracy and computational frugality represents a fundamental barrier; current high-fidelity causal models cannot be reliably deployed for real-time monitoring or edge-computing diagnostics in rapid-response systems.

3. The Lag Ambiguity Problem: The results mathematically demonstrate that continuous physical phenomena (such as pressure propagation) resist rigid temporal discretization. The non-parametric models successfully identified the existence of true causal links but struggled to pinpoint the exact discrete sensor lag, resulting in a massive +0.267 F1-Score improvement when shifting from strict directed evaluation (0.5333) to undirected evaluation (0.8000). This confirms that in physical systems, the causal footprint is temporally "smeared." Evaluating continuous physics through strict, discrete lag penalties artificially degrades model confidence; future operational deployments must evaluate causal influences over integrated time-windows.

The Causal Relevance Index for time series

We extended CRI to the temporal setting (CRI_{TS}) by replacing the i.i.d. assumption with stationarity and α-mixing, and proved that super-uniformity of p-values holds for RobustParCorr and GPDC under these conditions. Proposition 5.2 further showed that CRI remains a valid ranking criterion even under Type II errors, the default regime in time series. On the CausalChamber, CRI ranked GPDC above RobustParCorr above CMIknn, identifying the test with the strongest causal evidence despite RobustParCorr achieving higher precision; a distinction that F1 alone cannot make.

Practical Recommendations

1. **Match temporal resolution to the physics.** If the causal mechanism operates faster than the sampling rate, use PCMCI+ rather than PCMCI to avoid missing contemporaneous effects.
2. **Report undirected metrics alongside directed ones.** On physical systems, lag ambiguity can mask correct structural discoveries; undirected F1 improved by up to +0.27 in our experiments.
3. **Budget computation realistically.** CMIknn required over 3 hours on 2000 time steps. For real-time or resource-constrained deployments, a parametric pre-screen followed by targeted nonparametric testing may be the only viable strategy.
4. **Report CRI alongside F1.** CRI captures the *strength* of causal evidence without requiring a decision threshold α, and can distinguish methods that F1 ranks identically.
5. **Difference with caution.** First-differencing preserves fast effects (actuator $\rightarrow$ fan) but can destroy slow, integrated ones (fan $\rightarrow$ pressure).

Future Directions

Benchmarking and algorithms

Soft-lag evaluation. Metrics that assign partial credit at neighbouring lags (e.g., exponential decay: 50% at $|\Delta\tau|$=1, 25% at $|\Delta\tau|$=2) would better reflect continuous physical mechanisms than the current all-or-nothing directed scoring.

Hybrid parametric/nonparametric pipelines. RobustParCorr runs in seconds but misses nonlinear links; CMIknn captures them but is orders of magnitude slower. A two-stage approach using a fast parametric skeleton followed by targeted nonparametric refinement on uncertain edges could combine the strengths of both.

Broader physical testbeds. Our benchmark used a single CausalChamber configuration with four variables. Extending the evaluation to higher-dimensional setups, different physical regimes (e.g., turbulent vs. laminar flow), and other real-world testbeds would strengthen the generality of the findings.

Sensor resolution and lag ambiguity. Increasing the sampling frequency or developing algorithms that evaluate causal influence over integrated time windows rather than strict discrete lags could reduce the reality gap we observed.

CRI theory and extensions

CRI for model selection. Using CRI to compare candidate graphs across $\tau_{\max}$ or α settings requires characterising its finite-sample variance; bootstrap confidence intervals could make this practical without a held-out ground truth.

Non-stationarity without differencing. A sliding-window strategy estimating local CRI values could detect not only *which* edges exist but *when* they appear, avoiding the signal destruction caused by global differencing.

Theoretical guarantees for CMIknn. We excluded CMIknn from our super-uniformity claims due to the lack of asymptotic theory for knn estimators under mixing. Establishing formal conditions, or counterexamples, would close an important gap flagged in Appendix B.

Scaling to high dimensions. As d grows, the sparsity ceiling $\text{CRI}_{\text{TS}}^{\max} = (1+s)/2$ compresses toward 0.5, reducing discriminative range. A normalised or edge-local variant would extend applicability to large sensor networks.

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Appendix A
Strobl's framework

This appendix provides a rigorous analysis of the role played by the i.i.d. assumption in the PC-p framework of Strobl et al. [5]. We trace every point in the derivation where the assumption is invoked, explain what it guarantees, and identify what would need to replace it in a time-series setting. We then analyze, with the same level of detail, why the construction is structurally tied to the PC algorithm and cannot be applied as a black box to other causal discovery methods.

Precise role of the i.i.d. assumption

The i.i.d. assumption is *never explicitly named* as a standalone assumption in [5]. It is instead absorbed into the validity of the CI tests that the PC algorithm calls. We now identify every step in Strobl's derivation where this implicit assumption is operative.

Step 1: The CI test as a statistical procedure

Strobl's framework begins with a CI test for two random variables A and B given a conditioning set S. The test produces a test statistic $\hat{s}_i$ that estimates a population parameter s_i, and a p-value

$$p_i \;=\; \Pr\big(|\hat{s}_i| \geq |\hat{s}_i^{\text{obs}}| \;\big|\; s_i = 0\big), \tag{A.1}$$

where $s_i = 0$ encodes the null hypothesis that $A \perp\!\!\!\perp B \mid S$.

For this p-value to be *valid*—meaning $\Pr(p_i \leq \alpha \mid H_{0,i}) \leq \alpha$ for all $\alpha \in [0,1]$, i.e., super-uniformity—the test statistic $\hat{s}_i$ must have a known null distribution. This is where i.i.d. enters.

Example: Fisher's z-test. The most common CI test for continuous data computes the sample partial correlation $\hat{r}$ between A and B given S, then applies the Fisher z-transform:

$$z \;=\; \tfrac{1}{2}\ln\!\Big(\frac{1+\hat{r}}{1-\hat{r}}\Big)\cdot\sqrt{n-|S|-3}. \tag{A.2}$$

Under i.i.d. Gaussian sampling, the asymptotic theory gives:

$$z \;\xrightarrow{d}\; \mathcal{N}(0,1) \qquad \text{under } H_0 : \rho_{A,B|S} = 0, \tag{A.3}$$

where the convergence is in distribution as $n \to \infty$. The p-value is then $p_i = 2\big(1-\Phi(|z|)\big)$, which is exactly uniform on $[0,1]$ under H_0 in the Gaussian i.i.d. case, and asymptotically uniform for non-Gaussian i.i.d. data. The convergence (A.3) relies on two properties of i.i.d. samples:

(a) **The classical CLT.** The sample partial correlation $\hat{r}$ is a smooth function of sample covariances, each of which is an average of i.i.d. terms. The standard CLT gives $\sqrt{n}(\hat{r}-\rho) \xrightarrow{d} \mathcal{N}(0,\sigma^2_{\text{iid}})$, where σ^2_{iid} is the asymptotic variance computed under independence.

(b) **Known asymptotic variance.** Under i.i.d. sampling with $\rho = 0$, the asymptotic variance of $\sqrt{n}\,\hat{r}$ is exactly 1 (for the Fisher-transformed version). This is what allows the test to compare z against $\mathcal{N}(0,1)$ without estimating any additional nuisance parameters.

If the data is not i.i.d.—for instance, if it comes from a stationary time series with temporal dependence—then both (a) and (b) may fail:

- **Ibragimov's CLT for α-mixing sequences** [6, 7] must replace the classical CLT. The convergence $\sqrt{n}\,\hat{r} \xrightarrow{d} \mathcal{N}(0,\sigma^2_{\text{LR}})$ still holds under appropriate mixing conditions, but with a *different* variance σ^2_{LR}: the long-run variance

$$\sigma^2_{\text{LR}} \;=\; \sum_{k=-\infty}^{\infty} \text{Cov}\big(g(X_0), g(X_k)\big), \tag{A.4}$$

  where g is the influence function of the partial correlation.
- The true asymptotic variance of z_n under H_0 is $\frac{n-|S|-3}{n}\,\sigma^2_{\text{LR}}$, not 1, since the Fisher transform is asymptotically linear near $\rho = 0$. For positively autocorrelated processes, $\sigma^2_{\text{LR}} > \sigma^2_{\text{iid}} = 1$ because the cross-covariance terms contribute positively, so the test over-disperses relative to its reference distribution and is *conservative*: p-values are super-uniform and the Strobl bounds remain valid, though power is reduced. For negatively

autocorrelated processes, σ^2_{LR} can fall below 1, making the test anti-conservative; super-uniformity is not guaranteed in that case without an explicit variance correction.

- If the implementation corrects for autocorrelation by replacing n with an effective sample size $n_{\mathrm{eff}} < n$ estimated from the data, the p-values can be approximately uniform rather than merely super-uniform, restoring power while maintaining validity.

The full consequences of these cases for the two CI tests used in our experiments are established in Section B (Propositions B.1–B.2); we do not repeat that analysis here.

Example: the G-test. For discrete data, the G-statistic has a limiting χ^2 distribution with non-centrality parameter $g_i = 0$ under H_0. The same i.i.d. assumption underlies this limiting distribution: the G-statistic is a sum over i.i.d. cell counts. Under temporal dependence, the effective degrees of freedom change, but the same conservative/super-uniform argument applies if the reference distribution is not adjusted.

Step 2: The simplifying assumption on CI tests

Strobl's derivation of both bounds relies on an additional assumption stated informally on page 9 of [5] and used in every subsequent bound. We state it precisely.

Assumption A.1 (Oracle-test decoupling)**.** The probability that CI test i outputs a given result depends only on the output of CI oracle i, conditional on the outputs of all oracles. Formally, for any event E_i concerning the output of test i:

$$\Pr(E_i \mid \mathrm{oracle}_1, \ldots, \mathrm{oracle}_m) \;=\; \Pr(E_i \mid \mathrm{oracle}_i).$$

Strobl justifies this by noting that most CI test statistics have limiting distributions that depend only on their own null parameter. For instance, Fisher's z-statistic has a limiting $\mathcal{N}(0,1)$ that depends only on whether $\rho_{A,B|S} = 0$, not on the values of other partial correlations.

This assumption is reasonable under i.i.d. sampling because the limiting distributions of classical test statistics are determined entirely by their own null parameters. Under time-series dependence, the same data feeds into multiple CI tests, inducing correlations between test statistics. However, Assumption A.1 concerns only the *marginal null distribution* of each test—not the joint distribution across tests. In the time-series setting, PCMCI's regression step decorrelates the residuals before testing, supporting the marginal validity of each test even when the raw data is temporally dependent.

Step 3: The union bound (Strobl's Equation 5)

The union bound states that for m CI tests with the compound null $H_0 : \bigwedge_{i=1}^m (\mathrm{oracle}_i \text{ outputs independent})$:

$$\Pr(\text{Type I error}) \;=\; \Pr\Big(\bigcup_{i=1}^m \{\mathrm{test}_i \text{ outputs dependent}\} \;\Big|\; H_0\Big) \;\le\; \sum_{i=1}^m p_i. \tag{A.5}$$

This is Boole's inequality, a purely measure-theoretic result that holds for *any* probability measure, regardless of the dependence structure of the events. The only requirement from the data-generating process is that each p_i is super-uniform under its own null.

Conclusion: The union bound does not use i.i.d. It requires only super-uniform p-values.

Step 4: The intersection bound (Strobl's Equations 8–9)

The intersection bound states that for the compound null $H_0 : \bigvee_{i=1}^m (\mathrm{oracle}_i \text{ outputs independent})$:

$$\Pr(\text{Type I error}) \;=\; \Pr\Big(\bigcap_{i=1}^m \{\mathrm{test}_i \text{ outputs dependent}\} \;\Big|\; H_0\Big) \;\le\; \max_{j=1,\ldots,m} p_j. \tag{A.6}$$

The derivation proceeds as follows. Under H_0, at least one oracle i^* outputs "independent" (though we do not know which one). Then:

$$\Pr\Big(\bigcap_{i=1}^m \{\mathrm{test}_i \text{ outputs dependent}\} \;\Big|\; H_0\Big) \;\le\; \Pr\big(\mathrm{test}_{i^*} \text{ outputs dependent} \mid H_0\big) \tag{A.7}$$

$$= \Pr\big(\mathrm{test}_{i^*} \text{ outputs dependent} \mid \mathrm{oracle}_{i^*} \text{ outputs independent}\big) \tag{A.8}$$

$$= p_{i^*} \;\le\; \max_{j=1,\ldots,m} p_j. \tag{A.9}$$

Line (A.7) uses monotonicity of probability. Line (A.8) uses Assumption A.1. Line (A.9) uses super-uniformity of p_{i^*} and the fact that i^* is unknown, necessitating the maximum.

Conclusion: The intersection bound does not use i.i.d. It requires super-uniformity and Assumption A.1.

Step 5: The Benjamini–Yekutieli FDR procedure (Strobl's Equation 2)

Once edge-specific p-value bounds have been computed, Strobl feeds them into the BY estimator:

$$\widehat{\mathrm{FDR}}_{\mathrm{BY}}(\alpha) \;=\; \frac{m\,\alpha \sum_{i=1}^{m} i^{-1}}{\max\{R, 1\}}, \tag{A.10}$$

where m is the number of hypothesis tests and R is the number of rejections at level α.

Benjamini and Yekutieli (2001) [8] proved that (A.10) achieves strong FDR control under *arbitrary dependence among the m p-values*. BY is robust to inter-test dependence but not to intra-test invalidity: if a CI test produces anti-conservative p-values, BY loses its guarantee. The only requirement from the data is therefore, once more, super-uniformity of each individual p-value.

Conclusion: The BY procedure does not use i.i.d. in its handling of dependence among tests. It requires i.i.d. (or an equivalent condition) only to the extent that i.i.d. guarantees the validity of each individual p-value.

Step 6: Theorem 4 and the zero Type II error rate

Strobl's Theorem 4 states that, under three conditions, all p-value bounds in his Table 1 can be computed from the CI tests actually executed by PC:

(i) D-separation faithfulness.
(ii) Zero Type II error rate.
(iii) PC also tests non-adjacent pairs A, B with common neighbor C for conditional dependence given subsets containing C.

Condition (ii) is the one most affected by the data-generating process. Under i.i.d. sampling with large n, consistent CI tests have vanishing Type II error rates asymptotically. Under time-series dependence, the effective sample size is reduced due to autocorrelation, increasing the Type II error rate for a given nominal n. Power can be restored by using longer series, adjusting significance thresholds (Strobl recommends $\alpha = 0.20$ in practice), or using CI tests with better power for dependent data.

Importantly, as shown in Proposition 4.2, the CRI framework remains a valid ranking criterion even when Type II errors occur, provided the CI test has some power. The zero Type II error assumption is therefore sufficient but not necessary for the utility of CRI.

Summary: dependency chain of the i.i.d. assumption

We summarize the logical dependencies in the following chain:

$$\boxed{\text{i.i.d. data}} \xrightarrow{\text{classical CLT}} \boxed{\text{known null distribution of } \hat{s}_i} \xrightarrow{\text{calibration}} \boxed{\text{super-uniform } p_i}$$

$$\xrightarrow{\text{union/intersection bounds}} \boxed{\text{valid edge-specific p-value bounds}} \xrightarrow{\text{BY procedure}} \boxed{\text{FDR control}}$$

The i.i.d. assumption acts at the *first* link only. Everything downstream depends on super-uniformity of the p-values, not on i.i.d. directly. Any replacement for the first link that produces super-uniform p-values—such as **Ibragimov's CLT for α-mixing sequences** [6, 7] combined with either variance inflation or effective sample size correction—leaves the rest of the chain intact. This is precisely what we establish in Section B for RobustParCorr and GPDC.

Why the construction is specific to the PC algorithm

The super-uniformity discussion above explains what the data-generating process must guarantee. We now turn to what the algorithm must guarantee: that the Boolean formulas used to construct the edge-specific hypothesis tests are correct and that the algorithm performs all the CI tests needed to evaluate the bounds.

Reverse-engineering PC's decision logic

Strobl's approach is to write, for each edge type in the CPDAG, a Boolean formula expressing the edge's presence in terms of CI oracle outputs, then bound it using the union and intersection inequalities.

Skeleton edge $A - B$. By Proposition 1 of Spirtes et al. (2000) [**?**], under the global directed Markov property and d-separation faithfulness, $A - B$ is present if and only if

$$A \not\perp\!\!\!\perp B \mid S \qquad \text{for all } S \subseteq \mathrm{Pa}(A) \setminus B \text{ and all } S \subseteq \mathrm{Pa}(B) \setminus A. \tag{A.11}$$

The null "$A - B$ is absent" is a disjunction over CI oracles; the alternative is a conjunction. The intersection bound gives $p_{A-B} = \max_S p_{A \perp\!\!\!\perp B|S}$.
This formula relies on Proposition 1 and d-separation faithfulness. Score-based methods such as GES optimize a score function without performing CI tests, so the Boolean-formula approach is entirely inapplicable to them.

Unshielded v-structure $A \to C \leftarrow B$. Strobl's Proposition 3 characterizes the collider $A \to C \leftarrow B$ (with A, B non-adjacent) by conditional *dependence* of A and B given every subset of $\mathrm{Pa}(A) \setminus B$ containing C and every subset of $\mathrm{Pa}(B) \setminus A$ containing C. Combined with the skeleton conditions, the alternative is:

$$H_1 : \underbrace{(A - C)}_{\text{conjunction}} \wedge \underbrace{(B - C)}_{\text{conjunction}} \wedge \underbrace{\gamma_{AB|C}}_{\text{conjunction}} , \tag{A.12}$$

where $\gamma_{AB|C}$ denotes conditional dependence of A and B given all subsets containing C. The intersection bound gives:

$$\Pr(H_1 \mid H_0) \;\leq\; \max\{p_{A-C},\, p_{B-C},\, p_{\gamma_{AB|C}}\}.$$

This relies on Proposition 1 for the skeleton terms, Strobl's Proposition 3 for the v-structure characterization, and PC's specific separating-set logic (Algorithm 6). FCI uses a different v-structure detection procedure and would require different formulas.

Orientation rules. PC uses three deterministic Meek rules to orient additional edges after v-structure detection:

- Rule 1: $C \to A - B$ with C, B non-adjacent implies $A \to B$.
- Rule 2: $A \to C \to B$ with $A - B$ implies $A \to B$.
- Rule 3: $A - C \to B$, $A - D \to B$ with C, D non-adjacent implies $A \to B$.

For Rule 1, the Boolean formula is $H_1 : (A - B) \wedge (C \to A)$, giving $p_{A \to B} \leq \max\{p_{A-B}, p_{C \to A}\}$. When multiple justifications exist, the union bound sums over them: $p_{A \to B} \leq \max\{p_{A-B}, \sum_i p_{C_i \to A}\}$. These formulas are entirely determined by the Meek rules; a different algorithm with different orientation rules would require different formulas.

Theorem 4: PC must perform all necessary CI tests

The p-value bounds can only be computed if the algorithm actually performs all the CI tests that appear in the formulas. Strobl's Theorem 4 states that PC achieves this under d-separation faithfulness, zero Type II error rate, and condition (iii). The proof traces PC's execution path:

(i) **Skeleton.** PC tests $A \perp\!\!\!\perp B \mid S$ for all $S \subseteq N_{\hat{G}}(A) \setminus B$. Under zero Type II error, $N(A) \subseteq N_{\hat{G}}(A)$, so all necessary subsets are tested and p_{A-B} is computable.

(ii) **V-structures.** The additional tests with conditioning sets containing C (condition (iii)) ensure $p_{\gamma_{AB|C}}$ is computable.

(iii) **Orientation rules.** The bounds for rules 1–3 are recursive functions of the skeleton and v-structure bounds, already controlled.

PCMCI+ has a fundamentally different execution structure: it does not start from a fully connected graph, uses a pre-screening MCI phase to identify candidate parents, determines edge orientation from temporal ordering rather than v-structures and Meek rules, and constructs conditioning sets differently. Theorem 4's proof does not transfer to PCMCI+; an analogous result would require tracing PCMCI+'s execution path separately.

The PC-p modifications: stabilization and ambiguation

Strobl proposes two algorithmic modifications to mitigate Type II errors in practice.

Stabilization (from PC-stable). A false edge deletion can remove a variable from a neighborhood, preventing a later CI test from being executed with the correct conditioning set. PC-stable [**?**] freezes neighborhoods at each conditioning set size level, ensuring edge deletions within a level do not affect which tests are performed at the same level. This is specific to PC's progressive skeleton discovery strategy and does not apply directly to PCMCI+, though the underlying concern about Type II error propagation is equally relevant.

Ambiguation. When two v-structures conflict, PC-p unorients both and labels the edges as ambiguous, preventing orientation rules from propagating potential errors. This mechanism is designed around PC's three-stage pipeline and has no direct analogue in PCMCI+, where temporal ordering resolves most orientation ambiguities.

Decomposition: generic vs. PC-specific layers

Generic layer (portable to PCMCI+ and CRI).

(i) The union bound: $\Pr(\text{Type I}) \leq \sum_i h_i$. Requires only super-uniform h_i.
(ii) The intersection bound: $\Pr(\text{Type I}) \leq \max_j h_j$. Requires super-uniformity and Assumption A.1.
(iii) The max-aggregation for skeleton edges: $p_{A-B} = \max_S p_{A \perp\!\!\!\perp B|S}$. Requires faithfulness and the intersection bound.
(iv) The BY procedure for FDR control. Requires super-uniform p-values, handles arbitrary inter-test dependence.

PC-specific layer (not directly portable).

(i) Boolean formulas for v-structures, using Strobl's Proposition 3 and PC's separating set logic.
(ii) Boolean formulas for orientation rules 1–3, using the Meek rules.
(iii) Theorem 4's proof that PC performs all necessary CI tests.
(iv) The PC-stable stabilization and ambiguation modifications.

Implication for CRI. The CRI framework uses only the max-aggregated p-values $p''_e = \max_S p_e(S)$ from the generic layer. It does not require the v-structure or orientation rule bounds from the PC-specific layer, because PCMCI+ produces directed lag-indexed edges through its own temporal-ordering logic. CRI therefore inherits the full theoretical guarantees of the generic layer—Propositions 4.1 and 4.2, without requiring any PC-specific machinery.

Appendix B
Validity of CRI under time-series dependence

This appendix establishes rigorously that the Causal Relevance Index extends to time-series data produced by PCMCI+. We show that (1) the i.i.d. assumption in the original static framework can be replaced by stationarity and mixing, and (2) correlations among p-values, which inevitably arise in time series, do not compromise any of CRI's theoretical properties. We also provide the full super-uniformity proofs for the two CI tests for which we claim theoretical guarantees (RobustParCorr and GPDC).

Standing assumptions

We work under the following assumptions throughout this appendix and in the main text whenever Assumptions B.1–B.2 are invoked.

Assumption B.1 (Strict stationarity)**.** The process $(X_t)_{t\in\mathbb{Z}}$ is strictly stationary.

Assumption B.2 (Strong mixing)**.** The process (X_t) is α-mixing with mixing coefficients $\alpha(k) = O(k^{-\beta})$ for some $\beta > 2$.

These two assumptions are standard in the time-series causal discovery literature and are satisfied by a broad class of processes, including stationary ARMA, nonlinear autoregressive, and diffusion-type models. In particular, the CausalChamber wind tunnel data, which consists of smooth physical dynamics sampled at a fixed rate, falls comfortably within this class.

Remark B.1 (Simplifying assumption on CI tests)**.** The p-value bounds of Strobl et al. also rely on a third, often implicit, assumption: the null-hypothesis behavior of CI test i depends only on the output of CI oracle i, not on the other oracles (Assumption A.1 in Appendix A). This holds whenever the test statistic has a limiting distribution determined solely by its own null parameter (e.g., Fisher's $z \to \mathcal{N}(0,1)$ under $H_0 : z_i = 0$). In time series, the same data feeds into multiple CI tests, so test statistics can be correlated. However, the assumption concerns only the *marginal null distribution* of each test, not the joint distribution across tests. For PCMCI, the regression step decorrelates the residuals before testing, which supports this assumption under stationarity.

What CRI actually requires

We identify every mathematical property that the CRI definition and its theoretical guarantees (Propositions 4.1 and 4.2) rely on.

(R1) Super-uniformity of p-values under the null. For every truly absent edge e and at least one conditioning set S^* rendering the variables conditionally independent:

$$\Pr\big(p_e(S^*) \le u \mid H_{0,e}(S^*)\big) \;\le\; u \qquad \text{for all } u \in [0,1]. \tag{B.1}$$

(R2) Validity of the max-aggregation. The edge-wise p-value $p''_e = \max_{S\in\mathcal{S}_e^{\text{tested}}} p_e(S)$ satisfies $p''_e \ge p_e(S^*)$ for the separating set S^* above.

(R3) Linearity of expectation. The CRI is defined as $\mathrm{CRI}_{\mathrm{TS}}(G) = 1 - \frac{1}{|\mathcal{E}|}\sum_{e\in\mathcal{E}} p''_e$, and all theoretical properties are derived by taking expectations of this average.

We now verify that none of these requirements depends on i.i.d. data or on independence among p-values.

Requirement (R1): replacing i.i.d. with stationarity and mixing

Where i.i.d. enters in the original framework

In Strobl et al. [5], condition (B.1) is guaranteed by the known null distributions of Fisher's z-test and the G-test under i.i.d. sampling. For Fisher's z-test:

$$z \;=\; \tfrac{1}{2}\ln\!\Big(\frac{1+\hat r}{1-\hat r}\Big)\cdot\sqrt{n-|S|-3}, \tag{B.2}$$

where $\hat r$ is the sample partial correlation. Under i.i.d. Gaussian data and $H_0 : \rho = 0$, the classical CLT gives $z \xrightarrow{d} \mathcal{N}(0,1)$, and the p-value $p = 2(1-\Phi(|z|))$ is exactly uniform on $[0,1]$ under H_0. This relies on:

(a) The classical CLT applied to sample covariances, yielding asymptotic normality of $\hat r$.
(b) The asymptotic variance of $\sqrt{n}\,\hat r$ being exactly 1 under H_0, so no variance estimation is needed.

What changes under time-series dependence

Under a stationary α-mixing process (Assumptions B.1–B.2), property (a) is replaced by **Ibragimov's CLT for α-mixing sequences** [6, 7]: the sample partial correlation is still asymptotically normal, but with a different variance:

$$\sqrt{n}\,\hat{r} \xrightarrow{d} \mathcal{N}(0,\, \sigma^2_{\mathrm{LR}}), \tag{B.3}$$

where $\sigma^2_{\mathrm{LR}} = \sum_{k=-\infty}^{\infty} \mathrm{Cov}(g(X_0), g(X_k))$ is the long-run variance. Property (b) is modified: $\sigma^2_{\mathrm{LR}} \neq 1$ in general. The true asymptotic variance of z under H_0 is therefore $\frac{n-|S|-3}{n}\,\sigma^2_{\mathrm{LR}}$, not 1. Two cases arise:

- If $\sigma^2_{\mathrm{LR}} > 1$ (the case for positively autocorrelated processes): the test rejects less often than α, so p-values are super-uniform. Condition **(R1)** is automatically satisfied.
- If the implementation uses $n_{\mathrm{eff}} < n$ (as `tigramite` does): the correction targets σ^2_{LR}, yielding approximately uniform p-values. Any under-correction preserves super-uniformity.

Conclusion for Requirement (R1)

The i.i.d. assumption can be replaced by stationarity and α-mixing. Ibragimov's CLT provides the same asymptotic normality as the classical CLT, and the resulting variance mismatch makes the test conservative rather than anti-conservative for positively autocorrelated processes. Super-uniformity is preserved. The detailed proofs for the two CI tests used in our experiments follow immediately below.

Super-uniformity proofs for individual CI tests

RobustParCorr

The RobustParCorr test, as implemented in `tigramite` [2], proceeds in three steps: (i) each variable's observations are mapped to approximate normality via the nonparanormal transform $Z^i_t = \Phi^{-1}\big(\hat{F}^i_n(X^i_t)\big)$, where $\hat{F}^i_n$ is the empirical CDF and Φ^{-1} the standard normal quantile function; (ii) the sample partial correlation $\hat{r}$ is computed from the transformed data; (iii) the Fisher z-statistic $z = \frac{1}{2}\ln\big(\frac{1+\hat{r}}{1-\hat{r}}\big)\sqrt{n_{\mathrm{eff}} - |S| - 3}$ is compared to $\mathcal{N}(0,1)$.

Proposition B.1 (Super-uniformity of RobustParCorr)**.** Under Assumptions B.1–B.2, the p-values produced by RobustParCorr are asymptotically super-uniform under the null hypothesis of conditional independence.

Proof. The proof has three steps.

Step 1: Uniform convergence of the empirical CDF under mixing. Let F^i denote the true marginal CDF of X^i_t (well-defined by stationarity). Under α-mixing with $\alpha(k) = O(k^{-\beta})$, $\beta > 2$, the Glivenko–Cantelli theorem for mixing sequences [9, 10] gives

$$\sup_x \big|\hat{F}^i_n(x) - F^i(x)\big| = O\Big(\sqrt{\frac{\log n}{n}}\Big) \quad \text{a.s.} \tag{B.4}$$

Step 2: Negligibility of the rank-transform estimation error. Write the population-level transform as $Z^{i,*}_t = \Phi^{-1}(F^i(X^i_t))$ and the estimated transform as $\hat{Z}^i_t = \Phi^{-1}(\hat{F}^i_n(X^i_t))$. On any compact subset $[\varepsilon, 1-\varepsilon]$, the quantile function Φ^{-1} is Lipschitz with constant L_ε. Using a standard winsorization argument at $\varepsilon_n = (\log n)/n$ [11], the boundary effects are controlled, giving

$$\max_{t=1,\dots,n} \big|\hat{Z}^i_t - Z^{i,*}_t\big| = O\Big(\frac{(\log n)^{3/2}}{\sqrt{n}}\Big) \quad \text{a.s.} \tag{B.5}$$

This perturbation affects the sample partial correlation at order $o(n^{-1/2})$, so the asymptotic distribution of $\hat{r}_n$ computed from $\hat{Z}^i_t$ coincides with that computed from the (unobservable) population-transformed process $Z^{i,*}_t$.

Step 3: CLT for α-mixing sequences and variance comparison. The population-transformed process $\{Z^*_t\}$ is strictly stationary, α-mixing with the same coefficients as $\{X_t\}$ (since α-mixing is preserved under measurable transformations), and has standard normal marginals by construction.

Under H_0 (conditional independence), the population partial correlation is $\rho = 0$. By the CLT for α-mixing sequences (Ibragimov, 1962; Ibragimov and Linnik, 1971) [6, 7], provided $\alpha(k) = O(k^{-\beta})$ with $\beta > 2$ and $E[|g(Z^*_0)|^{2+\delta}] < \infty$ for some $\delta > 0$:

$$\sqrt{n}\,\hat{r}_n \xrightarrow{d} \mathcal{N}(0,\, \sigma^2_{\mathrm{LR}}),$$

where $\sigma^2_{\mathrm{LR}} = \sum_{k=-\infty}^{\infty} \mathrm{Cov}\big(g(Z^*_0), g(Z^*_k)\big)$ is the long-run variance and g is the influence function of the partial correlation functional.

The Fisher z-test compares z_n to $\mathcal{N}(0,1)$, implicitly assuming variance 1. Under H_0 with $\rho = 0$, the Fisher z-transform is asymptotically linear ($\frac{1}{2}\ln\!\left(\frac{1+\hat{r}}{1-\hat{r}}\right) \approx \hat{r}$), so the true asymptotic variance of z_n is $\frac{n_{\text{eff}}-|S|-3}{n}\,\sigma^2_{\text{LR}}$. Super-uniformity holds whenever this quantity is ≥ 1. Two cases arise:

- *No effective sample size correction* ($n_{\text{eff}} = n$). For positively autocorrelated processes, the cross-covariance terms in σ^2_{LR} are non-negative, so $\sigma^2_{\text{LR}} \geq \sigma^2_{\text{iid}} = 1$ and the test is conservative, hence p-values are super-uniform. This covers the typical case in physical systems such as the CausalChamber wind tunnel data. For negatively autocorrelated processes, σ^2_{LR} can fall below 1, making the test anti-conservative; super-uniformity is not guaranteed in that case without an explicit variance correction.
- *With effective sample size correction* ($n_{\text{eff}} < n$, as in `tigramite`). The correction estimates σ^2_{LR} and reduces n_{eff} accordingly. If accurate, p-values are approximately uniform; if the correction slightly underestimates σ^2_{LR} (the typical finite-sample behavior), p-values remain super-uniform.

In both cases, the p-values are asymptotically super-uniform under H_0 for the class of positively autocorrelated processes considered here. □

Remark B.2. The moment condition $E[|g(Z_0^*)|^{2+\delta}] < \infty$ is easily satisfied: after the rank transform, the data has standard normal marginals (up to boundary effects controlled by winsorization), which have moments of all orders.

GPDC

The Gaussian Process Distance Correlation test (GPDC) [12] works as follows: (i) Gaussian process regressions of each variable on the conditioning set S are performed, producing residuals $\hat{\varepsilon}^i_t$ and $\hat{\varepsilon}^j_t$; (ii) the distance correlation $\mathrm{dCor}(\hat{\varepsilon}^i, \hat{\varepsilon}^j)$ is computed; (iii) the p-value is obtained by comparing the observed distance correlation to an analytic gamma approximation [13].

Proposition B.2 (Super-uniformity of GPDC)**.** Under Assumptions B.1–B.2, and provided the GP regression model is correctly specified (i.e., the true conditional mean lies in or can be consistently approximated by the reproducing kernel Hilbert space of the GP kernel), the p-values produced by GPDC are asymptotically super-uniform under the null.

Proof. **Step 1: Residual decorrelation.** Under H_0, the population residuals $\varepsilon^i_t = X^i_{t-\tau} - E[X^i_{t-\tau} \mid S]$ and $\varepsilon^j_t = X^j_t - E[X^j_t \mid S]$ are independent. Under correct GP specification, $\hat{\varepsilon}_t \to \varepsilon_t$ as $n \to \infty$. Moreover, because the conditioning set S typically includes lagged values that capture temporal structure, the GP regression removes the predictable component, leaving residuals that are approximately temporally uncorrelated.

Step 2: Null distribution of distance correlation. Under H_0 with independent observations, $n \cdot \mathrm{dCor}^2$ converges to a weighted sum of χ^2_1 random variables [14], well approximated by a gamma distribution. Since the GP residuals are approximately i.i.d. under H_0 (by Step 1), the same limiting distribution applies. Any residual temporal correlation inflates the variance of the distance correlation statistic relative to the i.i.d. reference, making the gamma approximation conservative. The test therefore rejects less often than α, and the p-values are super-uniform. □

Remark B.3. The GPDC argument is less sharp than the RobustParCorr argument because it relies on the quality of the GP regression. In practice, GPDC tends to be conservative—the gamma approximation over-estimates the null distribution's tail, which further supports super-uniformity. For the CausalChamber wind tunnel dataset, the squared exponential kernel provides a reasonable approximation to the smooth physical dynamics.

Remark B.4 (On CMIknn)**.** The third CI test used in our experiments, CMIknn [12], estimates conditional mutual information via k-nearest neighbors and uses a local permutation scheme for the null distribution. A clean theoretical guarantee of super-uniformity under mixing is not currently available: the knn estimator's bias and variance under temporal dependence are not well characterized, and the local permutation scheme provides only empirical calibration. We therefore include CMIknn as an experimental comparison but do not claim theoretical super-uniformity for it. Extending the formal guarantees to CMIknn is an interesting direction for future work.

Requirement (R2): the max-aggregation does not need i.i.d.

The max-aggregation $p''_e = \max_S p_e(S)$ is a deterministic function of the individual p-values, applied pointwise for each realization. Its validity as an upper bound on the p-value of the intersection hypothesis test follows from the intersection bound (Strobl's Equation 9):

$$\Pr\!\big(\text{all tests reject} \mid H_0\big) \;\leq\; p_{i^*} \;\leq\; \max_j p_j, \tag{B.6}$$

where i^* is the index for which the oracle outputs "independent" under H_0. This bound uses only:

(a) Monotonicity of probability: $\Pr(\bigcap_i E_i) \leq \Pr(E_{i^*})$.
(b) Assumption A.1 (Appendix A): the null distribution of test i depends only on oracle i's output.
(c) Super-uniformity of p_{i^*} (Requirement **(R1)**).

None of these involve the joint distribution of $(p_1, \ldots, p_m)$. The max-aggregation is therefore valid regardless of any correlations among the p-values $p_e(S)$ for different conditioning sets S.

Requirement (R3): why correlated p-values are not a problem for CRI

The distinction between CRI and FDR control

Strobl et al. use the edge-wise p-values p''_e for FDR control via BY, which requires super-uniformity of each p-value plus the $\sum i^{-1}$ correction factor to handle arbitrary inter-test dependence. BY is designed for *thresholding*: the dependence structure among p-values directly affects how conservative the cutoff is.

CRI does something fundamentally different: it *averages* the p-values into a single score without thresholding, without selecting edges, and without multiple testing. All its theoretical properties follow from a single tool: **linearity of expectation**.

Linearity of expectation is unconditional

Proposition B.3 (CRI properties are dependence-free)**.** Let $p''_1, \ldots, p''_m$ be random variables taking values in $[0, 1]$. Assume only that for every truly absent edge e: $\mathbb{E}[p''_e \mid H_{0,e}] \geq \frac{1}{2}$, which follows from Proposition 4.1*(ii)*. Then all properties in Propositions 4.1–4.2 hold with no assumption on the joint distribution of $(p''_1, \ldots, p''_m)$.

Proof. **Proposition 4.1*(i)*: false-positive control.** $\Pr(p''_e \leq \alpha) \leq \Pr(p_e(S^*) \leq \alpha) \leq \alpha$ follows from $p''_e \geq p_e(S^*)$ and super-uniformity of $p_e(S^*)$. This is a marginal statement; no joint distribution is involved.

Proposition 4.1*(ii)*: null baseline. This is established in Proposition 4.1*(ii)* and is not repeated here.

Proposition 4.1*(iii)* and Proposition 4.2*(ii)–(iii)*. These involve $\bar{p} = \frac{1}{|\mathcal{E}|} \sum_e p''_e$ and its expectation. By linearity of expectation:

$$\mathbb{E}[\bar{p}] \;=\; \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \mathbb{E}[p''_e]. \tag{B.7}$$

This identity holds for any joint distribution of $(p''_1, \ldots, p''_m)$ — it requires no independence, uncorrelatedness, or any other structural assumption. The proofs of Propositions 4.1*(iii)* and 4.2*(ii)–(iii)* decompose the sum into truly absent and truly present edges and apply the marginal bounds $\mathbb{E}[p''_e] \geq \frac{1}{2}$ (absent edges) and $\mathbb{E}[p''_e] < \frac{1}{2}$ (present edges with sufficient power). At no point does the dependence structure among the p''_e enter the argument. □

Why this is stronger than what Strobl needs

Property needed	Strobl (FDR)	CRI
Super-uniformity of each p''_e	Yes	Yes
Dependence correction (BY's $\sum i^{-1}$)	Yes	**No**
Joint distribution of $(p''_1, \ldots, p''_m)$	Matters	**Irrelevant**
Thresholding / multiple testing	Yes	No

CRI's requirements are strictly weaker than those of FDR control because averaging depends only on marginal expectations, not on the joint structure.

Sources of correlation among p-values in time series

The sources of correlation among p-values in the time-series setting are identified in Remark B.1 (shared data, overlapping conditioning sets, temporal autocorrelation, PCMCI+'s iterative conditioning). As established by Proposition B.3, none of them affect CRI's theoretical guarantees: the first three are marginal properties already handled by super-uniformity, and the fourth affects only the joint structure of the p''_e, which linearity of expectation does not require.

Summary

CRI's extension to time series is valid under stationarity and α-mixing:

1. The i.i.d. assumption enters Strobl's framework only through super-uniformity of CI test p-values. Under stationarity and mixing, **Ibragimov's CLT for α-mixing sequences** [6, 7] replaces the classical CLT, and the resulting variance inflation makes tests conservative for positively autocorrelated processes, preserving super-uniformity (Propositions B.1–B.2).
2. The max-aggregation $p''_e = \max_S p_e(S)$ is valid regardless of dependence among p-values, relying only on monotonicity of probability and marginal super-uniformity.
3. All theoretical properties of CRI follow from linearity of expectation, which holds unconditionally (Proposition B.3). Correlations among p-values are irrelevant to CRI's guarantees.

CRI therefore inherits its full theoretical properties—null baseline, false-positive control, true-positive equivalence, and the plausibility ranking of Proposition 4.2—in the time-series setting without any additional assumptions beyond stationarity and mixing.

Appendix C
Theoretical Foundations

C.1 Causality and Structural Causal Models

The unprecedented success of modern machine learning and statistical modeling largely stems from their ability to identify complex patterns in high-dimensional data. However, these associational models suffer from a fundamental limitation: they are blind to the data-generating process. They capture correlation, but correlation is famously insufficient for decision-making. To understand why causal discovery is necessary, we must first examine the critical failures of purely associational reasoning.

Spurious Correlations

Standard statistical models estimate the conditional probability $P(Y \mid X)$, capturing how the observation of variable X updates our belief about variable Y. However, a high statistical dependence does not imply that manipulating X will change Y. Two variables may exhibit strong correlation entirely due to a shared, unobserved common cause (a confounder). For instance, the robust correlation between ice cream sales and shark attacks does not mean restricting ice cream sales will save swimmers; both are spuriously correlated via a common cause: warm weather. Associational models cannot distinguish between predictive features and manipulative causes.

Simpson's Paradox

The most striking demonstration of the limits of purely observational data is Simpson's Paradox. This phenomenon occurs when a statistical trend appears within specific, partitioned groups of data but completely vanishes or reverses when the groups are combined.

In a medical context, a drug might appear harmful to the general population but beneficial to both men and women when analyzed separately. Without a theoretical framework mapping how gender, drug assignment, and recovery influence one another, it is mathematically impossible to know whether the aggregated or partitioned data represents the true effect of the drug. Causality resolves this by determining the correct covariates to adjust for based on the underlying structural graph.

The Ladder of Causation

To formalize the boundaries of statistical learning, Judea Pearl introduced the Ladder of Causation, an epistemological hierarchy categorizing the levels of queries a system can answer.

- **Level 1: Association (Seeing)**: This level relies entirely on observational data to answer questions of the form, "What is the expected value of Y given that I observe X?" It is formalized as $P(Y \mid X)$. All standard machine learning—no matter how deep the neural network—operates exclusively on this level.
- **Level 2: Intervention (Doing)**: This level asks, "What will happen to Y if I force X to take a specific value?" This is denoted by Pearl's do-operator: $P(Y \mid do(X = x))$. Interventions represent active manipulation of the environment, predicting the results of actions rather than passive observations.
- **Level 3: Counterfactuals (Imagining)**: The highest level involves retrospective reasoning: "Given that Y was y and X was x, what would Y have been had X been x'?" This is mathematically represented as $P(Y_{X=x'} \mid X = x, Y = y)$.

Causal discovery is the necessary scientific process that allows us to construct the models required to climb from Level 1 (observational data) to Levels 2 and 3 (interventional and counterfactual reasoning).

Epistemological and Mathematical Foundations

To discover causal relationships from purely observational data, we must establish a bridge between statistical properties (independencies in the data) and causal properties (edges in a graph). This bridge is built upon several foundational assumptions.

Reichenbach's Common Cause Principle

The philosophical bedrock of causal discovery is Reichenbach's Common Cause Principle. It dictates that if two random variables X and Y are statistically dependent, there must be a causal explanation: either X causes Y, Y causes X, or there exists a third variable Z that causes both. Importantly, if we condition on the common cause Z, the variables X and Y become statistically independent.

The Causal Markov Condition

The Causal Markov Condition establishes the link between a causal Directed Acyclic Graph (DAG) and the joint probability distribution of the data. It states that every variable in a causal graph is conditionally independent of all its non-descendants, given its direct causes (its parents). Let PA_i denote the parents of variable X_i. The

condition implies that the joint distribution factorizes over the graph:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid PA_i)$$

This condition allows us to look at a proposed causal graph and read off the statistical independencies we should find in the data.

The Faithfulness Assumption

While the Markov condition implies that a missing causal link results in statistical independence, the Faithfulness assumption provides the converse: if we observe a statistical independence in our data, it is strictly due to the absence of a causal connection, not due to an accidental cancellation of parameters. It assumes that nature is not "tricking" us by having two opposing causal paths exactly offset one another.

Causal Sufficiency

A common, albeit strong, assumption in basic causal discovery algorithms is Causal Sufficiency. This assumes that there are no unobserved confounders in the system; every common cause of any two or more variables in the dataset is also measured and included in the dataset. Relaxing this assumption is a major focus of advanced causal discovery methods.

Structural Causal Models (SCMs)

An SCM is formally defined as a tuple $M = \langle U, V, F \rangle$:

- **U (Exogenous Variables)**: These are background variables whose values are determined outside the model. They represent unmeasured noise, individual variations, or external factors that introduce randomness into the system.
- **V (Endogenous Variables)**: These are the observable variables within our system. Their values are entirely determined by the exogenous variables in U and other endogenous variables in V.
- **F (Structural Equations)**: A set of functions $\{f_1, f_2, \ldots, f_n\}$ corresponding to the variables in V. Each equation describes how a specific variable $X_i \in V$ is generated by its direct causal parents PA_i and its specific exogenous noise U_i:

$$X_i = f_i(PA_i, U_i)$$

Graph Surgery and the do-operator

The true power of an SCM lies in its ability to mathematically define interventions. In an SCM, performing an intervention $do(X_i = x)$ corresponds to "graph surgery." It means we reach into the model, remove the structural equation f_i that normally governs X_i, and replace it with the constant x.

Visually, this removes all incoming causal edges to X_i in the DAG, creating a new, "mutilated" graph and a new interventional probability distribution. Causal Discovery, therefore, is the vital inverse problem: observing the data generated by the joint distribution and attempting to systematically reverse-engineer the underlying DAG and the structural equations F that created it.

C.2 Causality in Time Series

How Time Changes the Rules of Causality

In standard cross-sectional data, we assume data points are independent and identically distributed (i.i.d.). Time series data violently breaks this assumption. However, time also gives us the most powerful heuristic in causal discovery: the arrow of time.

- **Breaking the Symmetry**: In static data, if X and Y are dependent, it is mathematically difficult to tell if X $\rightarrow$ Y or Y $\rightarrow$ X without making strict assumptions about the noise. In time series, a cause must strictly precede or be simultaneous with its effect. The future cannot cause the past.
- **Autocorrelation**: Variables depend on their own past. If we do not account for the fact that Y_{t-1} causes Y_t, we might falsely conclude that X_{t-1} causes Y_t just because X and Y are correlated over time.

Granger Causality: The Classical Baseline (and its Limits)

You must include Granger Causality in your thesis because it is the historical baseline for time series analysis. However, framing it critically through the lens of Pearl's framework will score you major points for theoretical depth.

The Concept: Introduced by Clive Granger (1969), the core idea is based on predictability. Variable X is said to "Granger-cause" variable Y if past values of X contain information that helps predict Y_t better than using the past values of Y alone.

The Mathematical Formulation: Typically modeled using Vector Autoregression (VAR). If we are predicting Y_t:

$$Y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Y_{t-i} + \sum_{i=1}^{p} \beta_i X_{t-i} + \epsilon_t$$

If the coefficients β_i are jointly significantly different from zero, then X Granger-causes Y.

Granger causality is different from causality in the structural SCM sense.

- **It is still Level 1**: Granger causality is purely predictive (associational). It looks at $P(Y_t \mid Y_{<t}, X_{<t})$. It does not simulate interventions ($do(X)$).
- **Vulnerability to Confounding**: If an unobserved variable Z_{t-1} causes both X_t and Y_{t+1}, Granger tests will falsely claim X causes Y simply because X reacts to the confounder faster than Y does.

Structural Causal Models over Time

To find true causality in time series, we extend the Structural Causal Model (SCM) framework to dynamic environments. We unfold the Causal DAG across time, creating a Full Time Graph.

In a time series SCM, we observe a multivariate time series $V = \{V_t^1, V_t^2, \ldots, V_t^d\}$ for $t \in \mathbb{Z}$. The structural equations become:

$$V_t^j = f_j(PA(V_t^j), U_t^j)$$

Where the parents $PA(V_t^j)$ can now come from two distinct temporal dimensions:

- **Lagged Effects (Cross-temporal)**: Causes that occur in previous time steps (e.g., $X_{t-2} \to Y_t$).
- **Contemporaneous Effects (Instantaneous)**: Causes that occur within the same measured time step (e.g., $X_t \to Y_t$). This often happens when the underlying causal mechanism operates faster than our sampling rate (e.g., financial algorithms trading in milliseconds, but data is recorded in minutes).

Core Assumptions for Time Series Causal Discovery

Just like static data relies on the Markov condition and Faithfulness, time series causal discovery requires temporal assumptions to be solvable:

- **Causal Sufficiency** (or explicit handling of latent variables): Just as before, assuming no unmeasured common causes.
- **Causal Stationarity** (Time-Homogeneity): This is a massive theoretical assumption. It states that the causal structure and the mechanisms (the structural equations f_j) do not change over time. The rule that governs how X_{t-1} affects Y_t is the exact same rule that governs how X_{t-5} affects Y_{t-4}. Without this, we cannot pool data across time to test for conditional independencies.
- **No Backward-in-Time Edges**: By the laws of physics, a variable V_{t+k} cannot be in the parent set $PA(V_t^j)$ for any $k > 0$.

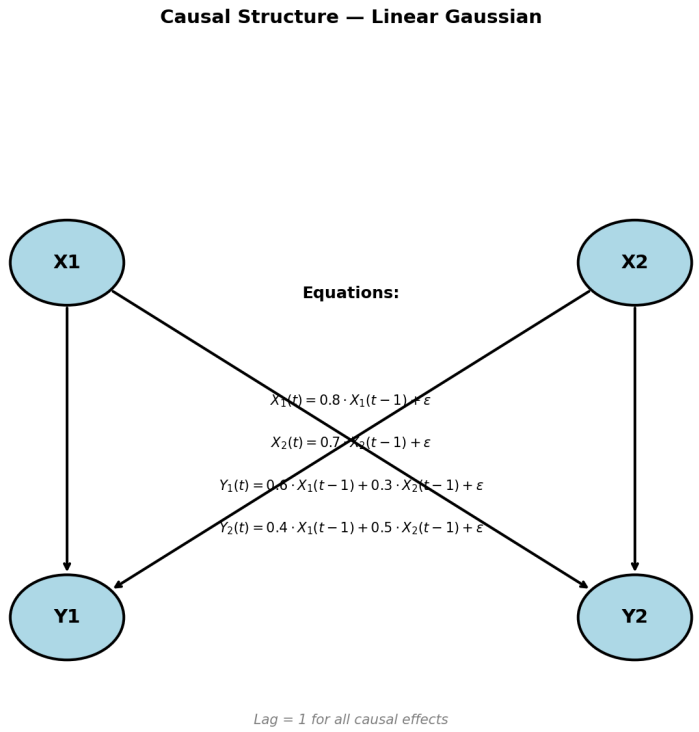

Figure C.1: Linear Gaussian Causal Structure

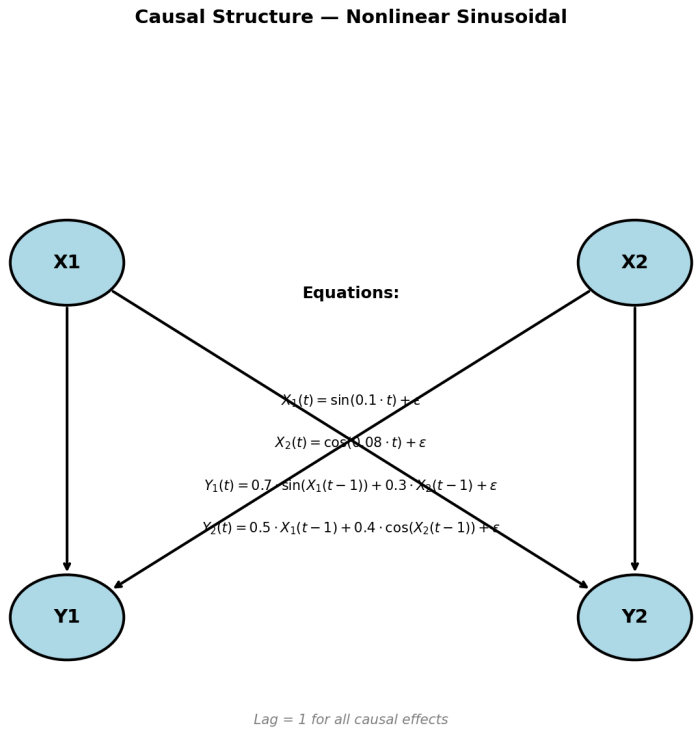

Figure C.2: Nonlinear Sinusoidal Causal Structure

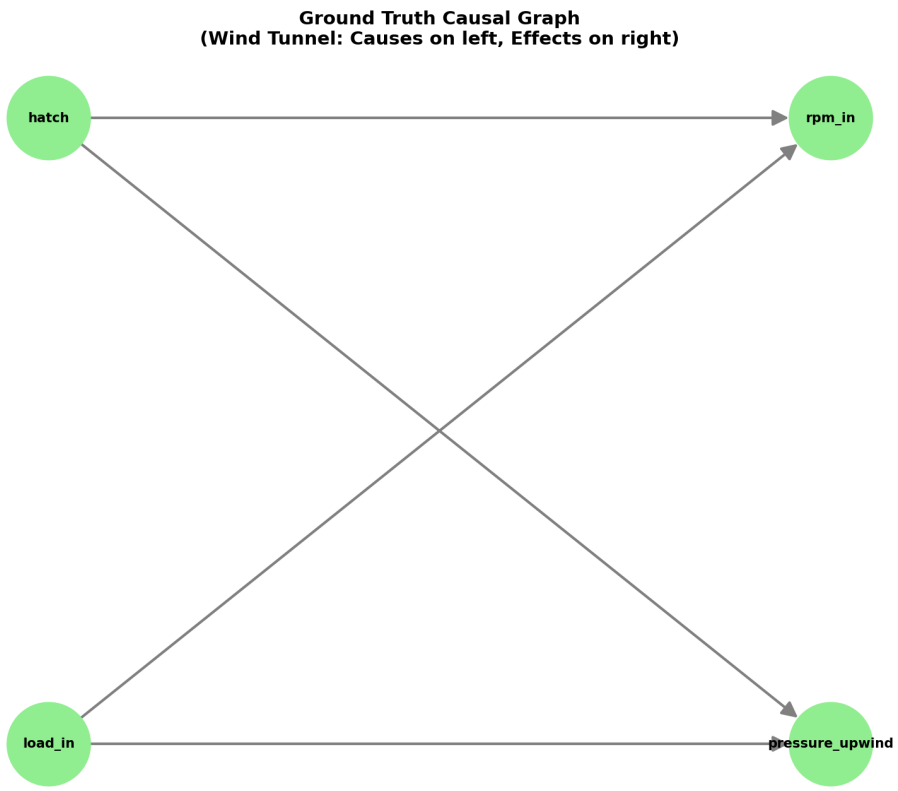

Figure C.3: Ground Truth Causal Graph for the CausalChamber physical wind tunnel experiment. The diagram isolates the four core interacting variables, showing the directed spatiotemporal dynamics from the manipulated causes (hatch, load_in) to the measured effects (rpm_in, pressure_upwind).

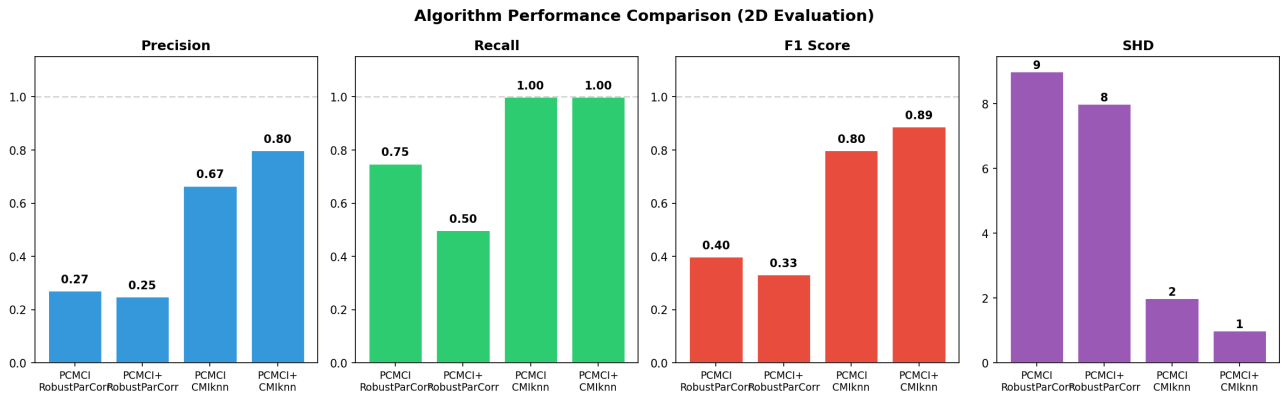

Figure C.4: Algorithm Performance Comparison (2D Evaluation). This chart details the Precision, Recall, F1 Score, and Structural Hamming Distance (SHD) across the four evaluated causal discovery methods.

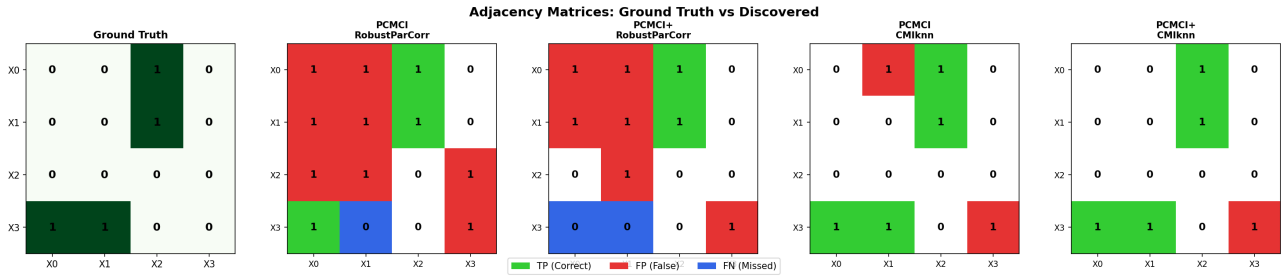

Figure C.5: Adjacency Matrices: Ground Truth vs. Discovered. This visual comparison maps the true causal links against the links discovered by each algorithm, highlighting True Positives (green), False Positives (red), and False Negatives (blue).